Genome Biology and Evolution, 2026, pp. 1–29
10.1093/gbe/evae000
Methods

RhizAeon: Fast, Accurate, Flexible and Scalable Recombination Analysis

Sergei L. Kosakovsky Pond,[1,2,*] Danielle Callan,[1,2] Steven Weaver,[1] Hannah Verdonk,[1,2] Jordan Zehr,[1,2] Spyros Lytras,[3] Taylor K. Paisie,[4] Maciej F. Boni,[1,2] Anton Nekrutenko[5] and Darren P. Martin[4]

[1]Institute for Genomics and Evolutionary Medicine, Temple University, Philadelphia, PA, USA
[2]Department of Biology, Temple University, Philadelphia, PA, USA
[3]Antigen Evolution and Design Lab, Department of Structural Biology and Chemistry, Institut Pasteur, Paris, France
[4]Institute of Infectious Disease and Molecular Medicine, University of Cape Town, Cape Town, South Africa
[5]Department of Biochemistry and Molecular Biology, The Pennsylvania State University, University Park, PA, USA

[*]Corresponding author: spond@temple.edu

Abstract

Homologous recombination transfers genetic material between genomes and obscures the evolutionary relationships among biological taxa. Because recombination distorts phylogenetic inference, existing methods either partition whole alignments into recombination-free segments or strip recombined tracts from individual mosaic genomes. However, the most widely used tools face severe computational bottlenecks as datasets grow: whole-alignment partitioning methods such as GARD scale exponentially with breakpoint count and cubically with the number of genomes, while triplet-scanning methods such as 3SEQ scale cubically with genome count. Here we introduce `RhizAeon`, an open-source framework that detects and characterizes recombination events in linear time by tracking continuous trajectories across genetic distance embeddings. Across simulated benchmarks designed to test common failure modes of recombination detectors, `RhizAeon` maintains false-positive rates below 2%, matches the overall detection power of 3SEQ, and more accurately resolves complex mosaic genomes with more than two breakpoints or three or more parents. In empirical analyses of Sarbecovirus genomes, multidrug-resistant *Streptococcus pneumoniae* chromosomes (2.22 Mb), and primate *HLA* Class I genes, `RhizAeon` recovers published whole-genome recombination patterns in seconds, resolving transfers from < 100 to $> 70{,}000$ nucleotides. `RhizAeon` is available both as a command-line tool and as an interactive WebAssembly web application.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Ancestral Recombination Graph

Significance Statement

Homologous recombination and horizontal gene transfer reshuffle DNA and RNA across viruses, bacteria, and eukaryotes, violating the single-tree assumption of standard phylogenetics. Existing recombination detection tools scale either exponentially with breakpoint count or cubically with genome count, creating a major bottleneck for modern genomic surveillance. We present `RhizAeon`, an open-source mathematical framework and software platform that reformulates recombination detection as continuous trajectory tracking across genetic distance spaces in linear time. `RhizAeon` maps whole-genome recombination across hundreds of viral, bacterial, and eukaryotic genomes in seconds, both from the command line and inside any web browser.

1. Introduction

Homologous recombination—hereafter referred to simply as recombination—drives genetic diversification across most viruses and cellular organisms. Why recombination is so widespread remains debated, though three mechanisms likely dominate. First, it repairs double-stranded DNA breaks in cellular organisms. Second, it purges harmful mutations to reverse maladaptive evolution. Third, it combines distinct beneficial mutations within a single genome to accelerate adaptation (Simon-Lorière and Holmes, 2011; Didelot and Maiden, 2010). Notably, many RNA viruses actively promote homologous recombination even though it plays no role in repairing broken RNA strands, suggesting that its primary adaptive value often lies in shuffling mutations across generations. Recombination rates vary widely across viruses (Simon-Lorière and Holmes, 2011) and cellular lifeforms (Didelot and Maiden, 2010; McVean et al., 2004), and breakpoints cluster in specific genomic hotspots (McVean et al., 2004; Lytras et al., 2022). Depending on where breakpoints fall, recombination can yield offspring that are either more or less fit than their parents (Simon-Lorière and Holmes, 2011; Lytras et al., 2022). In pathogenic microbes, recombination can alter transmission dynamics,

© The Author(s) 2026. Copyright and licence statements to be updated by the publisher during production.

cellular tropism, host range, pathogenicity, antigenicity, immune evasion, and antimicrobial resistance (Alfonsi et al., 2024; Didelot and Maiden, 2010; Croucher et al., 2011). In sexually reproducing eukaryotes, meiotic recombination purges deleterious mutations across generations and assembles hybrid genomes with novel phenotypes (Rieseberg et al., 2003; McVean et al., 2004).

Beyond its biological role, recombination confounds phylogenetic analyses. Standard phylogenetics assumes that homologous sequences share a single bifurcating tree. When a recombinant genome combines sequence tracts from divergent parents, forcing it onto a single tree shortens deep internal branches, lengthens terminal branches, and creates spurious clades (Schierup and Hein, 2000; Kosakovsky Pond et al., 2006). These distortions bias molecular clock estimates, node dating, ancestral sequence reconstruction, transmission cluster inference, and demographic history (Schierup and Hein, 2000; Posada, 2002). In codon-based selection analyses, recombination-induced homoplasies mimic convergent amino acid substitutions across branches, causing standard tests such as MEME and BUSTED (Murrell et al., 2012, 2015) to generate false-positive signals of diversifying selection (Kosakovsky Pond et al., 2006).

Consequently, molecular evolution pipelines routinely screen alignments to locate recombination breakpoints and identify recombinant sequences. Whole-alignment partitioning methods such as GARD (Kosakovsky Pond et al., 2006) locate disruptive breakpoints across an alignment and split it into non-recombinant segments for downstream phylogenetic inference. When breakpoints are so dense that partitioned segments become too short for reliable tree inference, "triplet-scanning" methods such as 3SEQ (Boni et al., 2007; Lam et al., 2018) and RDP5 (Martin et al., 2021) test every three-sequence combination to localize breakpoints within individual genomes, allowing researchers to mask or remove recombined tracts. These two strategies reflect a deeper algorithmic divide. GARD formulates breakpoint detection as a phylogenetic model selection problem under continuous-time Markov substitution models, using likelihood ratio tests to optimize breakpoint positions and segment trees. Because the search space of partition boundaries and tree topologies scales exponentially with the number of breakpoints B and cubically with the number of taxa N ($\mathcal{O}(L^B \cdot N^3)$ for alignment length L), multi-breakpoint GARD searches across hundreds of genomes require days of cluster computing.

In contrast, triplet-scanning methods such as 3SEQ and RDP5 avoid expensive phylogenetic likelihood optimizations and instead scan sequence triplets for localized shifts in pairwise sequence similarity. Although fast on alignments of a few hundred sequences, exhaustive triplet scanners scale cubically with taxon count ($\mathcal{O}(N^3L)$) and become impractical above a few thousand genomes. Moreover, because an isolated sequence triplet lacks a broader phylogenetic reference frame, triplet scanners cannot inherently distinguish the recombinant sequence from its parents—especially when close relatives of the parents are unsampled. Triplet scanners therefore require downstream phylogenetic post-processing to assign parent and recombinant roles and to consolidate redundant calls. Without consolidation, a single ancestral recombination event can trigger tens or hundreds of separate triplet alerts. Finally, without phylogenetic context, triplet scanners can mistake lineage-specific bursts of substitution in a genomic region (locus- or site-specific heterotachy) for recombination. Adding these downstream phylogenetic checks substantially increases total runtime beyond the initial triplet scan. Here we introduce `RhizAeon`, a fast, open-source recombination detection framework that scales linearly with dataset size while preserving phylogenetic context without building discrete trees. Its name combines the Greek *rhiza* (root) and *aeon* (deep evolutionary time), evoking the interconnected biological rhizome in place of strictly bifurcating trees.

2. The RhizAeon Framework: Principles of Operation

`RhizAeon` models recombination through continuous trajectory mechanics and zero-drift force fields in linear time ($\mathcal{O}(NKL)$ for N taxa, alignment length L, and K dynamically selected reference sequences called "landmark" lineages). By measuring instantaneous sequence affinity against landmark lineages and routing lineage-specific rate variation into an unpolarized ancestral root sink, `RhizAeon` turns recombination detection into a trajectory deconvolution problem. Under clonal evolution, site-to-site fluctuations cancel out as a centered Brownian bridge. Recombination breaks this equilibrium and imparts sustained directional momentum, pinpointing breakpoint sites and grouping "co-recombinant" descendants of a shared ancestral recombinant into a single event. Scanning along an alignment from $5'$ to $3'$, a non-recombinant genome keeps a stable genetic identity: its closest relatives stay constant and trace parallel paths through sequence space. Recombination disrupts this stability by merging segments from two or more parental lineages. A single crossover has no intrinsic polarity: if a recombinant joins a segment from lineage A with a segment from lineage B, sites in the A segment treat A as the recipient background and B as the foreign donor, whereas sites in the B segment reverse those roles. For consistency, we define a sequence's *home* lineage (major parent) as the landmark with the highest cumulative affinity across the alignment, and its *donor* lineage (minor parent) as the landmark toward which affinity shifts within a localized tract. As a scan moves along a mosaic genome, the trajectory follows its home lineage and then shifts sharply toward a donor lineage across the recombined tract. To capture these shifts without building trees or sliding windows, `RhizAeon` evaluates the affinity of each query sequence against the landmark panel at every polymorphic site.

To compute site-level affinity, `RhizAeon` adapts the cross-attention operator from transformer neural networks into a deterministic, closed-form analytical formula that requires no neural network training or learned parameters. Each query genome is compared against the landmark keys at every polymorphic position. Rather than weighting all nucleotide matches equally, continuous-time Markov substitution probabilities weight each score: sharing a rare derived allele commands strong attention, whereas common transitions command less. To prevent private mutations from causing false matches, the landmark panel includes an unpolarized ancestral root token that absorbs alleles absent from all sampled landmarks. A softmax operator normalizes these scores into a probability distribution summing to one across all landmarks and the root token at each site. A zero-drift force field then converts these site-by-site attention weights into continuous trajectories. At each polymorphic site, attention toward a landmark acts as an attractive force pulling the query toward that landmark in metric space. Subtracting each landmark's alignment-wide mean attention creates a zero-drift force field where positive and negative pulls average to zero across the genome. Under clonal evolution, random mutations generate opposing pulls that cancel out like thermal Brownian motion, keeping the trajectory within its home

corridor. Inside a recombined tract, however, shared derived alleles exert a sustained pull toward the donor lineage, driving a macroscopic excursion away from the home corridor. The breakpoints correspond to the exact coordinates where this directional force turns on and off.

Different recombination architectures and donor sampling regimes produce distinct signatures within this force field. In a single-breakpoint crossover, a sequence starts in its home corridor and switches into a donor corridor at the breakpoint for the rest of the alignment. The directional force drives cumulative momentum up to a single peak at the breakpoint before returning to zero at the end of the sequence, forming a triangular arch whose vertex marks the breakpoint. In a double-breakpoint replacement (a cassette), the directional force turns on at the $5'$ breakpoint and turns off at the $3'$ breakpoint, forming a trapezoidal plateau with a steep climb at the first breakpoint, a flat top across the cassette, and a steep drop at the second breakpoint. When the true donor lineage is unsampled, the introgressed tract loses affinity to the home lineage without matching any sampled taxon. Rather than forcing a false match to a contemporary sequence or dropping the signal, `RhizAeon` detects the departure from the home lineage and labels the donor as `"Ghost"`. Investigators can then either retain the call as a basal or unsampled introgression or mask the tract before phylogenetic inference, without falsely implicating sampled background sequences. When a close relative of the unsampled donor is present in the cohort (a surrogate sister lineage), trajectory tracking detects the shared derived alleles inherited from their common ancestor and attributes the transfer to the sampled sister reference (achieving 85.0%–100.0% surrogate attribution accuracy across divergence tiers, and 100.0% at $d_{\text{parent}} \geq 0.05$; Section 3.2).

To localize breakpoints across tracts that span tens to tens of thousands of bases, `RhizAeon` pairs multi-scale force-field scanning with exact contingency-table polishing. Fixed sliding windows face a trade-off: narrow windows are sensitive to mutational noise, whereas wide windows blur short tracts. `RhizAeon` avoids this trade-off by combining recursive binary segmentation on the Brownian bridge with a bank of matched filters spanning 30 to 800 base pairs. Fine-scale filters capture short micro-conversions, while broad-scale filters and Brownian-bridge splits capture large transfers without distortion from local rate variation. Once the peaks of these multi-scale responses narrow candidate breakpoints to localized intervals, contingency-table polishing refines each boundary to single-nucleotide coordinates. Across each candidate transition zone, `RhizAeon` examines the sequence triplet (recombinant, home landmark, and donor landmark) at every polymorphic site where the home and donor differ. Using two-by-two Fisher's exact tests ($\alpha_{\text{split}} = 0.05/L$), the polisher recursively tests outward extensions and inward splits of the interval, trimming non-recombinant margins and snapping breakpoints to the exact informative sites that maximize donor-versus-home contrast. Where the parents share identical nucleotides across a junction, the polisher reports the exact uninformative interval $[c_{\text{left}}, c_{\text{right}}]$ between bounding informative sites rather than an arbitrary single base.

In heavily recombined cohorts, choosing reference landmarks only once risks including mosaic genomes in the reference panel, which can hide secondary transfers or invert parent and recombinant roles. To prevent landmark contamination without discarding sequences, `RhizAeon` uses a three-pass architecture: initial screening and graph adjudication (Pass 1), iterative forward scrubbing (Pass 2), and backward leave-one-in verification (Pass 3). In Pass 2, candidate recombinant tracts are ranked by signal-to-noise ratio and iteratively masked with gap characters in the working alignment. Outside its masked tract, each sequence remains intact and serves as a clean landmark reference. Gap-masked sites exert zero net force ($\Delta F = 0$) in the attention field, preventing artificial trajectory drift across masked intervals while preserving alignment indel distances elsewhere. Once Pass 2 converges to a recombination-free backbone, Pass 3 restores each candidate tract one at a time while keeping all other tracts masked. Re-evaluating and re-polishing each tract in isolation removes interference from neighboring events and detects parent-recombinant polarity inversions—cases where donor D initially appeared to be a recombinant between R and H. Whenever treating R as the recombinant yields sharper transition contrast and lower flanking strain, `RhizAeon` swaps the polarity, masks the tract in R, and restores parent D.

Before certifying any candidate event in Pass 3, an orthogonal three-stage audit checks the raw alignment sites, followed by ancestral event consolidation across the cohort. First, a Poisson informative-site floor requires at least three unambiguous informative sites inside the tract where the recombinant matches the donor allele rather than the home allele. Second, if a tract has a high density of mutations separating the query from its home landmark that are absent from all other sampled sequences, `RhizAeon` tests their spatial clustering: tight clusters from an unsampled outgroup are labeled as `"Ghost"` donor introgressions, whereas scattered private mutations are routed into the root sink and rejected. Third, for tracts attributed to a sampled donor, a two-by-two Fisher's exact test compares donor-matching versus home-matching polymorphic site counts inside the tract against the flanking regions (excluding monomorphic sites and sites where home and donor match) under cohort-wide family-wise error control. Together, iterative scrubbing, leave-one-in verification, and raw-site auditing keep false-positive rates low even under severe mutation rate heterogeneity and heterotachy. Finally, while sparse surveys often detect isolated singleton recombinants, viral outbreaks and bacterial clones frequently contain multiple descendants of a single recombinant ancestor. Graph adjudication clusters these co-recombinant sequences whenever they share compatible parents and overlapping breakpoint intervals, consolidating them into a single ancestral event. Because "home" and "donor" labels become arbitrary when a crossover spans roughly half the alignment, consolidation checks both direct ($\text{Home}_A \approx \text{Home}_B, \text{Donor}_A \approx \text{Donor}_B$ with overlapping tracts) and inverted ($\text{Home}_A \approx \text{Donor}_B, \text{Donor}_A \approx \text{Home}_B$ with complementary breakpoints) polarities so that co-recombinants with swapped major/minor parent labels are grouped into the same ancestral event.

3. Coalescent Simulation Benchmark and Operational Evaluation

3.1. Coalescent ARG Benchmark and Calibration Against 3SEQ

To benchmark `RhizAeon` against 3SEQ (Boni et al., 2007; Lam et al., 2018)—a fast, widely used exact triplet scanner—we evaluated both methods across the 3,200 coalescent simulation trials of Posada and Crandall (2001) ($N = 10$ taxa, $L = 1{,}000$ nt, Figure 1, Supplementary Table S1). Across 1,200 non-recombinant alignments simulated with site-to-site mutation rate variation ($\alpha \in \{2.0, 0.5, 0.05\}$), both methods held false-positive rates well below the nominal 5% threshold (Figure 1C). On these 10-taxon null alignments, 3SEQ recorded a 0.17% false-positive

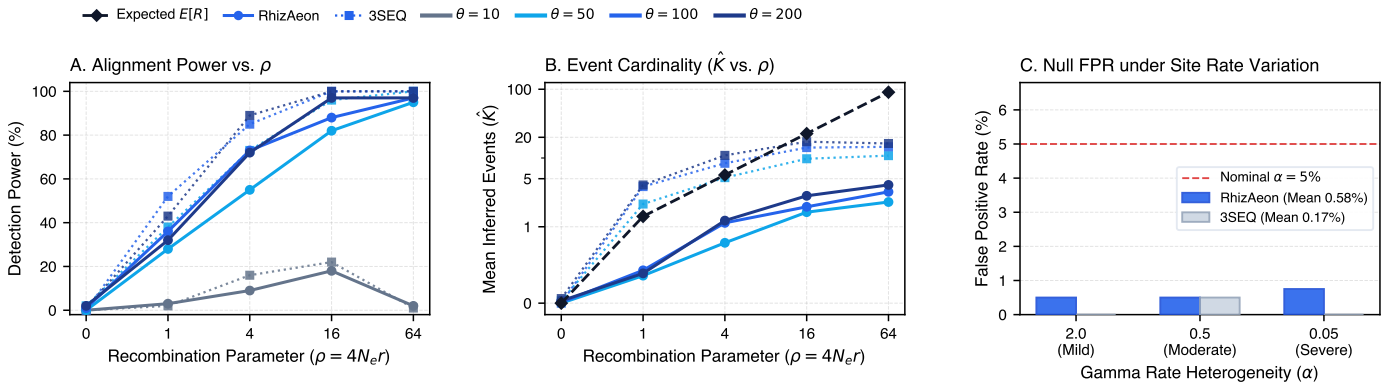


Figure 1 Performance of `RhizAeon` and 3SEQ across the 3,200-trial Posada and Crandall (2001) coalescent benchmark ($N = 10$, $L = 1{,}000$ nt). (A) Alignment-level detection power across population recombination rates ($\rho = 4N_e r \in [0, 64]$) and mutation densities ($\theta = 4N_e\mu \in [10, 200]$). (B) Inferred event cardinality ($\hat{K}$) versus Hudson's theoretical expectation ($\mathbb{E}[R]$, black dashed line), contrasting unconsolidated 3SEQ triplet counts against consolidated `RhizAeon` events. (C) Empirical false-positive rate (FPR, %) across 1,200 non-recombinant alignments under site-to-site Gamma rate variation ($\alpha \in \{2.0, 0.5, 0.05\}$; red dashed line marks the 5% nominal threshold).

rate (2 / 1,200 trials, 95% Wilson score interval [0.05%, 0.60%]), while **`RhizAeon`** recorded 0.58% (7 / 1,200 trials, 95% Wilson score interval [0.28%, 1.20%]), with 0.0% false positives across all 600 trials at $\theta \in \{10, 50\}$ and across 8 of the 12 rate-variation conditions. Occasional false positives occurred mainly at high sequence divergence ($\theta = 200$, accounting for 6 of the 7 false positives across $\alpha \in \{2.0, 0.5, 0.05\}$) due to random homoplasy across the 1,000 nt locus rather than extreme rate variation. Because the Posada and Crandall (2001) protocol varies mutation rates across sites rather than along individual branches, 3SEQ remains well-calibrated here. When lineage-specific rate accelerations occur (Section 3.3), private mutations on a single branch lack directional affinity toward contemporary donors and route harmlessly into **`RhizAeon`**'s root sink.

Across the 1,600 recombinant alignments ($\rho \in \{1, 4, 16, 64\}$), detection power depended primarily on mutation density (Figure 1A). In low-diversity datasets ($\theta = 10$, fewer than 25 variable sites), sparse mutations limited sensitivity: both **`RhizAeon`** and 3SEQ detected 1.0%–22.0% of alignments across $\rho \in [1, 64]$, where frequent crossovers split sequences into short segments carrying few mutations. At higher mutation densities ($\theta \geq 50$), 3SEQ's more permissive default triplet threshold gave it higher sensitivity at intermediate recombination rates (e.g., 73.0% vs. 55.0% at $\theta = 50, \rho = 4$), while both methods reached 88.0%–100.0% power at $\rho \geq 16$ ($\theta \in \{100, 200\}$). Even on small 10-taxon alignments with only $\binom{10}{3} = 120$ triplets, **`RhizAeon`** ran in a mean of 13.2–26.2 ms per alignment across $\theta \in [10, 200]$ (17.4 ms across all 1,200 rate-variation nulls), compared to 27.6–36.8 ms for 3SEQ. As cohort sizes grow to $N = 100$–$1{,}500$ taxa in the outbreak scaling benchmark below, **`RhizAeon`**'s $\mathcal{O}(NKL)$ architecture runs 73.8- to 224.0-fold faster than 3SEQ's $\mathcal{O}(N^3 L)$ triplet scan at $N = 100$–150 and finishes $N = 1{,}500$ in 28.1 s.

A key difference between **`RhizAeon`** and raw 3SEQ outputs is event count ($\hat{K}$) and bipartite event precision (Figure 1B and Supplementary Table S1). Because our Hudson coalescent simulator records the full Ancestral Recombination Graph (ARG) and extracts all observable topological transfers (K_{true}, Supplementary Section S7), we compared inferred event counts ($\hat{K}$) against both the true simulated ARG events and Hudson's theoretical expectation ($\mathbb{E}[R]$, dashed line). Without post-processing, 3SEQ outputs every sequence triplet with a significant hypergeometric excursion. Even when an alignment contains only one recombinant sequence, multiple combinations of proxy parent sequences—compounded by co-recombinant descendants sharing the same ancestral crossover—inflate raw triplet counts to $\hat{K} = 16.3$–17.3 at $\theta = 200, \rho \in \{16, 64\}$. In contrast, **`RhizAeon`** analyzes all sequences jointly via iterative scrubbing and leave-one-in verification, grouping shared segments into consolidated ancestral events ($\hat{K} \approx 1.23$–4.06 at $\theta = 200, \rho \geq 4$). Under one-to-one Hungarian bipartite matching weighted by parental triad attribution (S_{triad}) and SNP overlap (J_{SNP}, Materials and Methods, Equation 9), **`RhizAeon`** achieves 2.3- to 2.8-fold higher mean event-level precision than unconsolidated 3SEQ across $\theta \geq 50$ (36.5%–63.2% vs. 4.6%–46.5% at $\theta = 200$). However, as noted in the 3SEQ documentation (Boni et al., 2007; Lam et al., 2018), raw 3SEQ triplet lists are designed to be post-processed to remove redundant proxy-parent triplets and merge overlapping breakpoint intervals. We therefore implemented a downstream 3SEQ consolidation pipeline—**`3SEQ (Consolidated)`**—and compared it directly against both raw **`3SEQ`** and **`RhizAeon`** across four specialized benchmark suites.

3.2. Performance Across Specialized Surveillance and Evolutionary Regimes

Standard coalescent benchmarks test isolated crossovers in small, fully sampled trees, whereas real-world datasets rarely match these ideals (Figure 2). In natural populations, recombinant lineages expand into co-recombinant transmission clusters and acquire tracts ranging from $< 1\%$ to 50% of the genome from multiple or unsampled donors. To evaluate breakpoint localization, parent attribution, and cluster consolidation fairly, we compared **`RhizAeon`** against both unconsolidated **`3SEQ`** and post-processed **`3SEQ (Consolidated)`** across 2,280 alignments spanning four specialized regimes (Figure 2, Cols. 1–4).

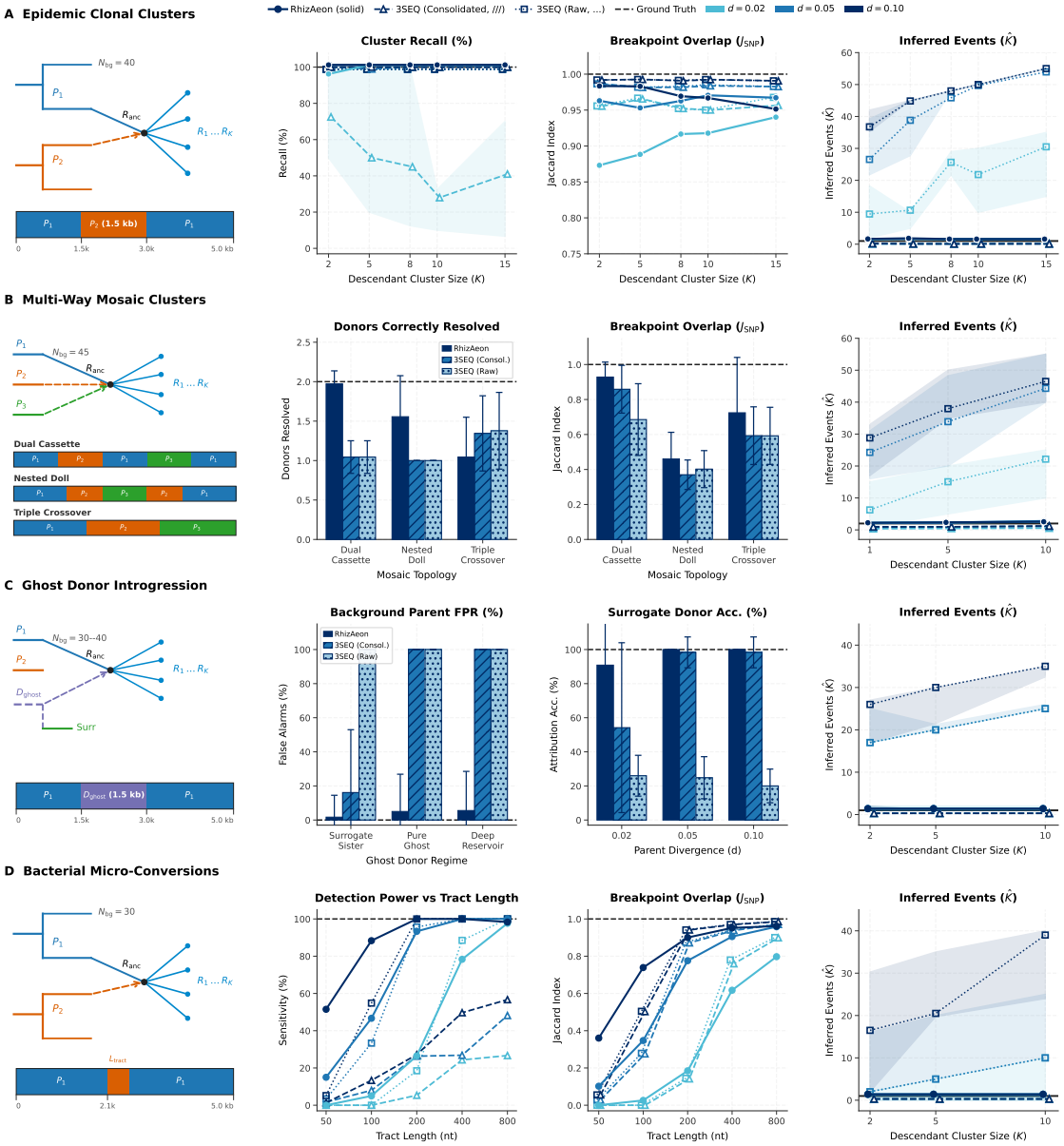


Figure 2 Comparison of `RhizAeon` (solid lines / dark navy bars), post-processed `3SEQ (Consolidated)` (dashed lines / hatched mid-blue bars), and unconsolidated `3SEQ (Raw)` (dotted lines / stippled light-blue bars) across four specialized genomic and epidemiological regimes ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$). Columns display the simulated topology schematic (Col. 1), primary detection, donor-attribution, or background FPR metric (Col. 2), breakpoint boundary overlap J_{SNP} or surrogate donor attribution accuracy (Col. 3), and inferred event cardinality $\hat{K}$ versus true ancestral event count K_{true} (Col. 4; overlapping curves at $\hat{K} = 1.0$ are slightly offset vertically for visibility). (A) Epidemic clonal expansion of a single ancestral 1.5 kb recombination cassette ($K_{\text{true}} = 1$) into $K_{\text{desc}} \in \{2\text{–}15\}$ sampled co-recombinant descendants across $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$. (B) Multi-way mosaic clusters combining two distinct donors ($D_{\text{true}} = 2.0$, $K_{\text{true}} = 2$) across Dual Cassette, Nested Doll, and Triple Crossover architectures. (C) Ghost donor introgression ($K_{\text{true}} = 1$) under Surrogate Sister, Pure Ghost, and Deep Reservoir sampling regimes, showing background parental false-positive rate (FPR, Col. 2), surrogate or ghost donor attribution accuracy (Col. 3), and inferred event cardinality $\hat{K}$ (Col. 4). (D) Bacterial micro-conversions ($K_{\text{true}} = 1$) across replacement tract lengths $L_{\text{tract}} \in [50, 800]$ nt.

Epidemic Clonal Expansion

During viral outbreaks and bacterial epidemics, an ancestral recombinant often spreads across hosts to form a cluster of K_{desc} sampled co-recombinant isolates (Figure 2A). Across 300 simulated outbreak cohorts (N = 42–55 taxa, L = 5,000 nt, $K_{\text{desc}} \in \{2, 5, 8, 10, 15\}$), unconsolidated `3SEQ` reported every significant triplet separately, inflating raw triplet counts from $\hat{K} = 9.5$–30.5 ($d_{\text{parent}} = 0.02$) to 36.8–55.0 ($d_{\text{parent}} = 0.10$) and flagging background parents via reciprocal triplet inversions. Downstream post-processing in `3SEQ (Consolidated)`

filters out reciprocal parent inversions using directional transition contrast and merges overlapping co-recombinant intervals. At moderate-to-high divergence ($d_{\text{parent}} \geq 0.05$), this consolidation resolved the single ancestral event (Figure 2A, Cols. 2–4), collapsing the cluster into $\hat{K} = 1.00$–1.10 events with 99.0%–100.0% recall, 93.8%–100.0% precision, $\leq 2.5\%$ background false positives, and 5.6–14.0 nt mean breakpoint error ($J = 0.98$–0.99). `RhizAeon` achieved comparable consolidation and accuracy directly in its linear-time pass ($\hat{K} = 1.00$–1.25 at $d_{\text{parent}} \geq 0.05$, and $\hat{K} = 1.00$ across 8 of 9 conditions for $K_{\text{desc}} \in \{8, 10, 15\}$ with $\hat{K} = 1.10$ at $d_{\text{parent}} = 0.02, K_{\text{desc}} = 15$; Figure 2A, Col. 4). It recovered 100.0% of co-recombinant descendants across all 10 conditions at $d_{\text{parent}} \geq 0.05$ (95.0%–100.0% at $d_{\text{parent}} = 0.02$, and 100.0% in 14 of 15 conditions overall) at 92.3%–100.0% precision (100.0% in 10 of 15 conditions) and 0.00% background false positives in 11 of 15 conditions. Mean breakpoint error was 12.4–39.3 nt at $d_{\text{parent}} = 0.10$ ($J = 0.95$–0.98) and 22.3–35.2 nt at $d_{\text{parent}} = 0.05$ ($J = 0.95$–0.97; Figure 2A, Col. 3), completing scans in 2.05–6.46 s versus 1.13–4.01 s for 3SEQ. At low parental divergence ($d_{\text{parent}} = 0.02$), `RhizAeon` recovered a much higher fraction of co-recombinant cluster members than `3SEQ (Consolidated)` as cluster size grew (100.0% recall vs. 28.0%–45.0% for $K_{\text{desc}} \in \{8, 10, 15\}$; dashed curves in Figure 2A, Col. 2).

Multi-Way Mosaic Clusters

Viral and bacterial recombinants often acquire segments from multiple donors, forming dual cassettes, nested "Russian doll" transfers, or serial triple crossovers (Figure 2B, Col. 1). By iteratively scrubbing detected segments and tracking excursions against all landmarks simultaneously, `RhizAeon` decomposes a single query genome into multiple donor transitions. In Dual Cassette recombinants (P_2 and P_3 cassettes in a P_1 background), `RhizAeon` recovered both donors in nearly every trial, averaging 1.97 of 2.00 expected donors overall (1.90–2.00) and 2.00 across all conditions at $d_{\text{parent}} \geq 0.05$ (Figure 2B, Col. 2). Across all Dual Cassette conditions, recall reached 100.0% with 97.1%–100.0% precision (100.0% in 7 of 9 conditions), $\leq 1.6\%$ background false positives (0.0% across all K_{desc} at $d_{\text{parent}} = 0.10$), and high boundary overlap ($J = 0.97$–0.98 at $d_{\text{parent}} = 0.10$, and $J = 0.95$–0.98 across $d_{\text{parent}} \geq 0.05$; Figure 2B, Col. 3). In Nested Russian Dolls ($P_1 \to P_2 \to P_3 \to P_2 \to P_1$) and Triple Crossovers ($P_1 \to P_2 \to P_3$), `RhizAeon` resolved an average of 1.56 (1.40–1.75) and 1.04 (0.70–1.35) donors respectively while maintaining 95.0%–100.0% recall, 95.0%–100.0% precision, and $\leq 1.7\%$ background false positives (0.0% in 13 of 18 conditions). By contrast, 3SEQ's hypergeometric scan extracts only one dominant segment per triplet call, so even `3SEQ (Consolidated)` recovered only one donor per mosaic genome (1.00–1.15 of 2.00 donors in Dual Cassettes and 1.00 in Nested Dolls, Figure 2B, Col. 2). Moreover, when a mosaic carries tracts from two donors (P_2 and P_3), evaluating triplet (R, P_1, P_2) across the P_3 segment distorts the flanking baseline, causing `3SEQ (Consolidated)` to record 16.2%–32.9% background false positives in Dual Cassettes at $d_{\text{parent}} \geq 0.05$ (down from 47.1%–81.0% in raw 3SEQ, vs. 0.0%–0.11% for `RhizAeon` at $d_{\text{parent}} \geq 0.05$ and 0.0%–1.6% overall).

Ghost Donor Introgression

In empirical surveillance, donor lineages are frequently absent from the alignment, such as unsampled wildlife reservoirs or extinct variants (Figure 2C, Col. 1). In this regime, 3SEQ uses sampled sequences as proxy parents to detect similarity shifts and localize breakpoints. When a close relative of the unsampled donor is present (the Surrogate Sister regime), both `RhizAeon` and `3SEQ (Consolidated)` perform well. At $d_{\text{parent}} \geq 0.05$, `RhizAeon` achieved 100.0% recall, 100.0% precision, 0.0% background and parent false positives, $J = 0.86$–0.99 boundary overlap (0.96–0.99 in 5 of 6 conditions), and 100.0% surrogate sister attribution (84.0%–99.0% recall, 90.6%–96.3% precision, and 85.0%–100.0% attribution at $d_{\text{parent}} = 0.02$; Figure 2C, Cols. 2–3). Under the same conditions, `3SEQ (Consolidated)` achieved 100.0% recall, 88.7%–100.0% precision, $J = 0.97$–0.99, and 95.0%–100.0% surrogate sister attribution, improving substantially over raw 3SEQ (9.3%–39.3% precision and 37.5%–56.4% background false positives; Figure 2C, Cols. 2–3). However, when the donor branches basal to the alignment MRCA with no close sampled relative (the Pure Ghost and Deep Reservoir regimes), unrooted 3SEQ triplets lack an outgroup to orient which sequence diverged. A basal ghost tract in recombinant R (within clade P_1) makes R look like a distant outgroup inside the tract, while non-recombinant P_1 sequences stay close to P_1 inside the tract and to proxy P_2 outside it. Both raw `3SEQ` and `3SEQ (Consolidated)` therefore localize the breakpoints accurately ($J = 0.95$–0.99) but invert the recombinant identity: they miss R (0.0% recall for `3SEQ (Consolidated)`) and falsely flag non-recombinant P_1 leaves in 50.0% of background taxa (100.0% parent false-positive rate; Figure 2C, Col. 2). In contrast, `RhizAeon`'s iterative scrubbing and orphan-divergence audit flags tracts where a sequence departs sharply from its home clade without matching any sampled reference, labeling the donor as `"Ghost"`. Across Pure Ghost cohorts at $d_{\text{parent}} \geq 0.05$ ($K_{\text{desc}} \in \{2, 5, 10\}$), `RhizAeon` detects 85.0%–100.0% of ghost-introgressed recombinants (80.0%–96.5% at $d_{\text{parent}} = 0.02$) with 95.0%–100.0% precision (100.0% in 5 of 6 conditions at $d_{\text{parent}} \geq 0.05$), $J = 0.83$–0.99 boundary overlap, 85.0%–100.0% `"Ghost"` donor attribution, and 0.0% background false positives in 5 of 6 conditions (Figure 2C). Across Deep Reservoir cohorts ($d_{\text{reservoir}} = 0.15$), `RhizAeon` similarly achieves 80.0%–100.0% recall (90.0%–100.0% in 8 of 9 conditions) and 80.0%–100.0% `"Ghost"` attribution; at $d_{\text{parent}} \geq 0.05$, precision reaches 90.0%–100.0% (100.0% in 5 of 6 conditions; 76.8%–95.0% at $d_{\text{parent}} = 0.02$) with $J = 0.83$–0.98 boundary overlap and 0.0% background false positives in 5 of 6 conditions.

Bacterial Micro-Conversions and Large-Alignment Scaling

In bacterial transformation (e.g., *Streptococcus pneumoniae*, *Helicobacter pylori*), homologous replacement often transfers short "micro-conversion" tracts spanning $< 10\%$ of a locus (Figure 2D). Detecting micro-conversions depends on the number of distinguishing polymorphisms inside the tract. At $L_{\text{tract}} = 50$ nt ($L = 5{,}000$ nt), simulated tracts carried a mean of only 0.8–1.4 ($d_{\text{parent}} = 0.02$), 1.8–2.9 ($d_{\text{parent}} = 0.05$), and 4.9–5.2 ($d_{\text{parent}} = 0.10$) mutations. Because both `RhizAeon`'s matched filter and 3SEQ's hypergeometric scan enforce strict family-wise error control, neither method called recombination when 50 nt tracts carried fewer than two mutations ($d_{\text{parent}} = 0.02$; 0.0% recall and 0.0% false positives). As mutation counts rose to 2.8–5.2 SNPs at $L_{\text{tract}} = 50$ nt ($d_{\text{parent}} \in \{0.05, 0.10\}$), `RhizAeon` detected 15.0%–30.0% ($d_{\text{parent}} = 0.05, K_{\text{desc}} \in \{2, 5\}$) and 50.0%–54.5% ($d_{\text{parent}} = 0.10$) of micro-conversions with 100.0% precision and 0.0% background false positives, compared to 0.0%–10.5% for raw `3SEQ` and 0.0%–5.0% for `3SEQ (Consolidated)` (Figure 2D, Col. 2). At $L_{\text{tract}} = 100$ nt ($d_{\text{parent}} = 0.10$,

8.0–10.6 SNPs), **RhizAeon** reached 80.0%–100.0% recall (mean 88.3%) with 100.0% precision and 0.0% false positives, outperforming raw **3SEQ** (45.0%–70.5%) and **3SEQ (Consolidated)** (20.0%–45.0%). Across all 45 Micro-Conversion conditions ($L_{\text{tract}} \in [50, 800]$ nt), **RhizAeon** maintained 100.0% precision and 0.00% background false positives. For $L_{\text{tract}} \in \{200, 400, 800\}$ nt ($d_{\text{parent}} \geq 0.05$, ≥ 9.3 SNPs), **RhizAeon** achieved 85.0%–100.0% recall (100.0% in 15 of 18 conditions) with $\hat{K} = 0.85$–1.05 inferred events (1.00 in 14 of 18 conditions); mean summed two-boundary breakpoint error ($|\hat{u}_1 - u_1| + |\hat{u}_2 - u_2|$, or 7.3–14.1 nt per boundary) at $d_{\text{parent}} = 0.10$ was 18.5–28.0 nt at $L_{\text{tract}} = 200$ nt ($J = 0.88$–0.91), 15.6–28.1 nt at $L_{\text{tract}} = 400$ nt ($J = 0.94$–0.96), and 14.5–22.2 nt for $K_{\text{desc}} \in \{2, 10\}$ at $L_{\text{tract}} = 800$ nt ($J = 0.93$–0.98, Figure 2D, Col. 3). At $L_{\text{tract}} \in \{400, 800\}$ nt ($d_{\text{parent}} = 0.10$), post-processing in **3SEQ (Consolidated)** reduced raw 3SEQ triplet counts from $\hat{K} = 15.8$–40.0 (46.0%–99.8% background false positives) down to $\hat{K} = 1.00$–1.10 (0.0%–0.3% false positives, 10.5–13.5 nt error, $J = 0.97$–0.99), while **RhizAeon** achieved $\hat{K} = 0.95$–1.00, 0.0% false positives, and higher cluster recall (95.0%–100.0% vs. 20.0%–100.0%) in 0.37–2.03 s per alignment. Finally, across outbreak alignments scaling from $N = 50$ to $N = 1{,}500$ sequences ($L = 5{,}000$ nt, $d_{\text{parent}} = 0.05$, $K_{\text{desc}} \in \{10$–$150\}$), 3SEQ's runtime grew cubically from 2.96 s ($N = 50$) to 37.93 s ($N = 100$) and 160.62 s ($N = 150$) before exceeding the 240 s cutoff at $N \geq 250$ ($\mathcal{O}(N^3)$ extrapolated runtimes of ~ 12.4 min at $N = 250$, ~ 1.65 h at $N = 500$, ~ 13.2 h at $N = 1{,}000$, and ~ 44.6 h at $N = 1{,}500$). In contrast, **RhizAeon** finished $N = 100$ in 0.51 s (73.8-fold speedup), $N = 150$ in 0.72 s (224.0-fold speedup), $N = 250$ in 1.26 s, $N = 500$ in 3.24 s, $N = 1{,}000$ in 11.73 s, and $N = 1{,}500$ in 28.14 s, recovering 100.0% of recombinants at 100.0% precision in a single event ($\hat{K} = 1.0$) at every cohort size.

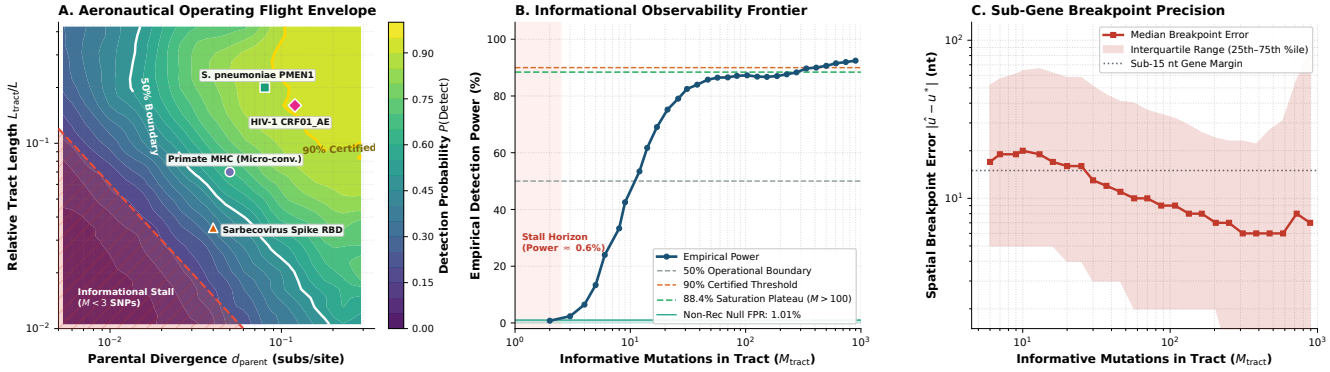


Figure 3 RhizAeon's operational phase envelope, observability frontier, and breakpoint precision across 1,000,000 continuous simulation trials ($N = 4$–250 taxa, $L = 1{,}000$–$30{,}000$ nt). (A) Detection power contours across parental divergence (d_{parent}) and relative tract length (L_{tract}/L), showing the 50% (white) and 90% (gold) boundaries, the informational stall zone ($M_{\text{tract}} < 3$ SNPs, red hatching), and empirical biological landmarks. (B) Detection power as a function of informative tract mutations (M_{tract}) relative to the non-recombinant null false-positive floor (1.01% alignment-level, 0.026% per-taxon, green band). (C) Median and interquartile breakpoint spatial error $|\hat{u} - u^*|$ across matched events versus M_{tract}.

3.3. The Continuous Operational Phase Envelope of Recombination Deconvolution

Just as an aircraft operating envelope defines safe flight boundaries across speed, altitude, and turbulence, an algorithmic operating envelope maps detection limits across continuous evolutionary space rather than a few discrete grid points. We mapped **RhizAeon**'s operational envelope across 1,000,000 independent simulation trials (Figure 3 and Supplementary Table S3), varying alignment length ($L \in [1{,}000, 30{,}000]$ nt), cohort size ($N \in [4, 250]$ taxa), parental divergence ($d_{\text{parent}} \in [0.005, 0.25]$ subs/site), and continuous Gamma rate heterogeneity ($\alpha \in [0.05, 5.0]$) alongside strict clock controls ($\alpha \geq 50$), spanning 333,334 clonal null alignments ($K_{\text{true}} = 0$) and 666,666 recombinant alignments with single crossovers, cassettes, and multi-donor mosaics ($K_{\text{true}} \in \{1, 2, 3, 4\}$). The sweep shows that recombination detection is governed not by relative tract length alone, but by the number of informative mutations separating donor and recipient inside the transferred tract ($M_{\text{tract}} \approx L_{\text{tract}} \times d_{\text{parent}}$, Figure 3A,B). With two or fewer informative mutations ($M_{\text{tract}} \leq 2$), power is 0.6% (484 / 85,564 trials, below the null false-positive floor) because fewer than three mutations cannot be distinguished from random Poisson clustering. Power rises to 7.0% at $M_{\text{tract}} \in [3, 5]$ (4,801 / 68,609), 32.3% at $M_{\text{tract}} \in [6, 10]$ (25,024 / 77,568), 56.4% at $M_{\text{tract}} \in [11, 15]$ (29,679 / 52,607), 72.9% at $M_{\text{tract}} \in [16, 25]$ (52,042 / 71,427), 83.3% at $M_{\text{tract}} \in [26, 50]$ (80,562 / 96,694), and 86.7% at $M_{\text{tract}} \in [51, 100]$ (73,443 / 84,676), plateauing at 88.4% when $M_{\text{tract}} > 100$ (114,518 / 129,521; 91.83% for $M_{\text{tract}} \geq 500$). The operational stall boundary forms a hyperbola in parameter space requiring $L_{\text{tract}} \times d_{\text{parent}} \geq 6$–$10$ informative SNPs (Figure 3A).

Across all 333,334 non-recombinant null alignments, **RhizAeon** held the alignment-level false-positive rate to 1.0062% (3,354 false positives), well below the nominal $\alpha = 5.0\%$ ceiling (Supplementary Figure S1A): 0.9134% (761 / 83,311) under a strict clock ($\alpha \geq 50$) and 1.0371% (2,593 / 250,023) under continuous Gamma heterotachy ($\alpha \in [0.05, 5.0]$), corresponding to a per-taxon false-positive rate of 0.0259% (5,884 / 22,744,905 leaves, 99.974% specificity) as private branch mutations routed into the root sink. Among matched events, median breakpoint error was 14.0 nt for single events ($K_{\text{true}} = 1$), 10.0 nt for double events ($K_{\text{true}} = 2$), and 8.0 nt for triple and quadruple events ($K_{\text{true}} \in \{3, 4\}$), sharpening to 7.0 nt for tracts with > 100 informative SNPs (8.0 nt for $M_{\text{tract}} \geq 500$; Figure 3C), while matched bipartite precision reached

94.5%, 96.7%, 97.5%, and 97.9% for $K_{\text{true}} = 1, 2, 3$, and 4 (Supplementary Figure S1B). Multi-donor mosaics ($P_1 \leftarrow P_2 + P_3$, $N = 179{,}233$) were resolved with 61.6% overall power (91.7% at $M_{\text{tract}} \geq 50$) and 0.488 event F_1 (0.794 at $M_{\text{tract}} \geq 50$), matching single-donor boundary overlap ($J = 0.882$ at $M_{\text{tract}} \geq 50$; Supplementary Figure S1D), and single-breakpoint crossovers showed higher sensitivity (68.2% across $N = 239{,}767$, median $J = 0.970$) than two-breakpoint cassettes (50.9% across $N = 426{,}899$, median $J = 0.944$; Supplementary Figure S1C). Across cluster nodes, all 1,000,000 trials completed with a median scan time of 885.0 ms per alignment. In practice, this envelope makes negative calls interpretable: above 15–25 informative SNPs (where sensitivity exceeds 73%–88

4. Empirical Genomic Applications

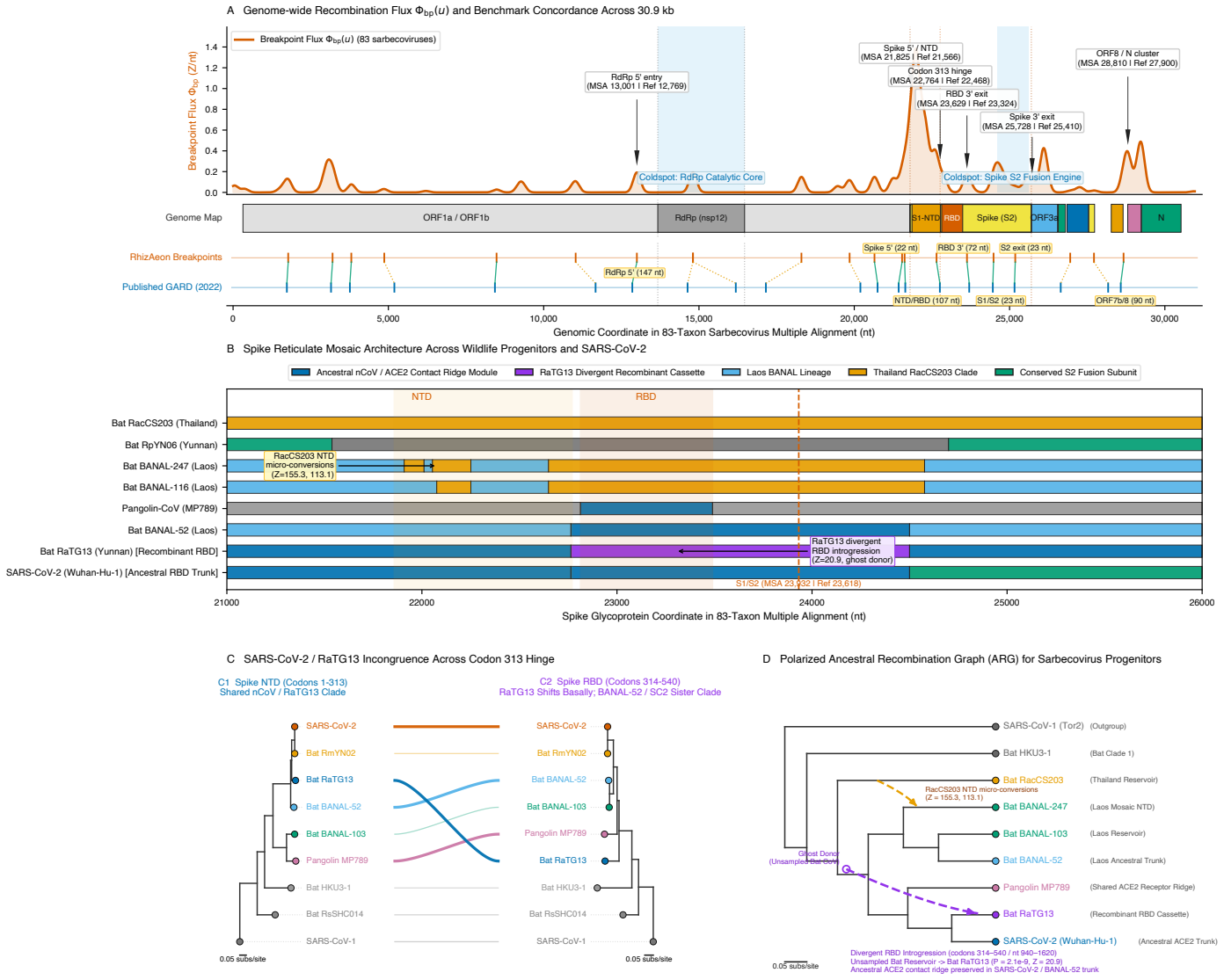


Figure 4 Sarbecovirus whole-genome recombination profiles, Spike mosaicism, and ancestral recombination graphs. (A) Genome-wide breakpoint flux $\Phi_{\text{bp}}(u)$ across 83 sarbecoviruses (30,965 nt MSA; Wuhan-Hu-1 reference NC_045512.2 coordinates in parentheses; Supplementary Table S2), aligned directly to the genomic ORF map above ($5'$–$3'$ MSA coordinate axis with vertical dotted guides at RdRp and Spike domain boundaries), highlighting hotspots across the S1 encoding region of Spike and ORF8/N, coldspots across RdRp and the S2 encoding region of Spike, and concordance between `RhizAeon` breakpoints (orange) and published GARD boundaries (blue, Lytras et al. 2022). (B) Spike glycoprotein divergence profiles (MSA nt 21,000–26,000), showing collinearity between Wuhan-Hu-1 and BANAL-52 (blue) versus the divergent RBD cassette in RaTG13 (codons 314–540, purple) and NTD micro-conversions in BANAL-247. (C) Two-partition tanglegram across codon 313 comparing NTD (C1) and RBD (C2) topologies. (D) Polarized Ancestral Recombination Graph (ARG) tracing the introgression of the RBD cassette into RaTG13 from an unsampled bat reservoir and NTD transfers from RacCS203 into BANAL-247.

4.1. Recombination Patterns in Sarbecoviruses

Sarbecoviruses—the coronavirus subgenus that includes SARS-CoV and SARS-CoV-2—recombine frequently in bat and other mammalian reservoirs. Screening these viral cohorts serves two goals: mapping genome-wide recombination hotspots and coldspots, and reconstructing the mosaic histories of individual viruses. Because exhaustive phylogenetic partitioning with GARD requires days of cluster computing on full 30 kb alignments, Lytras et al. (2022) had to downsample their 78-genome cohort to 22 representative viruses for GARD breakpoint screening. `RhizAeon` removes this bottleneck by projecting alignments into metric space and computing a continuous breakpoint flux $\Phi_{\mathrm{bp}}(u)$ in $\mathcal{O}(NKL)$ time. Without taxon subsampling, `RhizAeon` scanned the full 78-taxon alignment of Lytras et al. (2022) (30,956 bp MSA, 13,160 informative sites, identifying 37 consolidated events in 18.2 s) and an expanded 83-taxon cohort adding the five Laotian *Rhinolophus* BANAL isolates of Temmam et al. (2022) (30,965 bp MSA, 13,260 informative sites, identifying 45 consolidated events in 25.0 s). The resulting flux profile shows sharp regionalization across both the 83-taxon MSA and Wuhan-Hu-1 reference coordinates (NC_045512.2; Figure 4A and Supplementary Table S2). Pronounced coldspots span the catalytic core of RNA-dependent RNA polymerase (RdRp, 83-MSA nt 13,674–16,468, Ref nt 13,442–16,236) and the heptad-repeat fusion machinery of Spike S2 (83-MSA nt 24,600–25,600, Ref nt 24,283–25,283), where structural constraints limit modular exchange. Conversely, hotspots cluster at the Spike 5′ NTD junction (83-MSA 21,825 nt, Ref 21,566 nt), the codon 313 hinge between the NTD and receptor-binding domain (RBD, 83-MSA 22,764 nt, Ref 22,468 nt), the RBD 3′ exit (83-MSA 23,629 nt, Ref 23,324 nt, upstream of the S1/S2 furin site at 83-MSA 23,932 nt, Ref 23,618 nt), and the ORF8/Nucleocapsid boundary (83-MSA 28,810 nt, Ref 27,900 nt). Compared against the 21 published consensus GARD breakpoints of Lytras et al. (2022), `RhizAeon` breakpoints show a median deviation of 109.0 nt in Wuhan-Hu-1 reference coordinates on the 78-taxon cohort (12/21 within 150 nt, 57.1% concordance; 134.0 nt median in 78-MSA coordinates), improving in the 83-taxon cohort to 99.0 nt in reference coordinates (14/21 within 150 nt, 66.7% concordance) and 107.0 nt in 83-MSA coordinates (13/21 within 150 nt, 61.9% concordance; Figure 4A and Supplementary Table S2). Specifically, `RhizAeon` matched the Spike 5′ boundary within 22 nt (83-MSA 21,621 vs. 21,643 nt, Ref 21,389 vs. 21,411 nt), both the Spike S2 upstream and S2 exit boundaries within 23 nt (83-MSA 24,484 vs. 24,461 nt and 25,183 vs. 25,160 nt), the nsp2 junction within 43 nt (83-MSA 1,775 vs. 1,732 nt, Ref 1,723 vs. 1,680 nt), three nsp3 boundaries within 46–52 nt, the RBD 3′ exit within 72 nt (83-MSA 23,629 vs. 23,701 nt, Ref 23,324 vs. 23,396 nt), the ORF7a/ORF7b boundary within 19 nt in reference coordinates (Ref 27,369 vs. 27,388 nt), and the ORF7b/ORF8 boundary within 90 nt (83-MSA 28,670 vs. 28,580 nt, Ref 27,775 vs. 27,685 nt). Analyzing all 83 genomes also reveals fine-scale dynamics missed by downsampling: adding the Laotian BANAL isolates increases peak Spike NTD flux nearly four-fold (from 0.41 to 1.51 Z/nt) at the solvent-accessible antigenic supersite loops identified by cryo-EM, where bat viruses exchange micro-cassettes to evade antibodies without disrupting the spike trimer core (McCallum et al., 2021).

In parallel, trajectory tracking resolves the mosaic anatomy of individual lineages at single-nucleotide resolution (Figure 4B,C; throughout this section, Z_{phys} denotes the Brownian-bridge excursion signal-to-noise ratio and P_{Fisher} denotes the one-sided 2×2 Fisher's exact test across informative sites). In the Spike gene (3,843 nt alignment, 1,153 variable sites), `RhizAeon` pinpointed the canonical breakpoint at codon 313 ($Z_{\mathrm{phys}} = 8.23, P_{\mathrm{Fisher}} = 4.07 \times 10^{-22}$) and resolved the parent-recombinant ambiguity between SARS-CoV-2 and bat virus RaTG13 (Boni et al., 2020). In early three-way comparisons that lacked Laotian bat isolates, discordance between a RaTG13-like NTD and a pangolin-like ACE2-binding motif prompted speculation that SARS-CoV-2 arose from a recent recombination event (Li et al., 2020; Wrobel et al., 2020). However, trajectory tracking shows unbroken collinearity between Wuhan-Hu-1 and Laotian bat virus BANAL-52 across the entire Spike gene ($d = 0.0556$ in NTD, $d = 0.0617$ in RBD, and $d = 0.0457$ in S2; Figure 4B), confirming that the human-compatible ACE2 contact ridge is an ancestral trait circulating in Southeast Asian *Rhinolophus* bats. Conversely, `RhizAeon` identifies RaTG13 as the recombinant: while sharing the ancestral NTD and S2 backbone ($d \approx 0.05$), its divergence jumps over 2.5-fold inside the RBD ($d = 0.1380$; Figure 4C), reflecting an RBD import from an unsampled bat lineage (Boni et al., 2020). Multi-scale scanning also detects three recombination cassettes in Laotian bat virus BANAL-247 (Figure 4B): while its Spike backbone matches northern Laos *Rhinolophus malayanus* viruses (> 98.5% identity to BANAL-236/103), its NTD carries two micro-cassettes at 83-MSA nt 21,909–22,011 (Ref 21,647–21,749 nt, $Z_{\mathrm{phys}} = 155.3$) and 83-MSA nt 22,054–22,251 (Ref 21,791–21,983 nt, $Z_{\mathrm{phys}} = 113.1$), plus a third cassette across the NTD/RBD junction (83-MSA nt 22,318–22,592, Ref 22,044–22,297 nt, $Z_{\mathrm{phys}} = 81.2$), all donated by a Thai *Rhinolophus shameli* lineage related to RacCS203 (Figure 4B,D; Wacharapluesadee et al. 2021).

`RhizAeon` combines these transfers into a polarized Ancestral Recombination Graph (ARG; Figure 4D). In Spike, model selection across 9 sarbecoviruses favors a tripartite modular ARG (H_3: NTD chassis, divergent RBD cassette, and S2 backbone, $\Delta\mathrm{BIC} = 0.0$) over unpolarized trees ($\Delta\mathrm{BIC} = +170.9$) and a clonal null ($\Delta\mathrm{BIC} = +1654.8$), confirming that RaTG13 acquired its RBD from an unsampled bat reservoir ($Z_{\mathrm{phys}} = 20.9, P_{\mathrm{Fisher}} = 2.1 \times 10^{-9}$) while the ancestral SARS-CoV-2 / BANAL-52 trunk retained the ACE2 contact ridge (94.2% synapomorphic concordance, $P_{\mathrm{Fisher}} = 1.75 \times 10^{-23}$) and cut residual metric strain by 74.0% (3.34% → 0.87%). Across the full 30.9 kb alignment of 83 sarbecoviruses, consolidating the 45 detected events across 9 genomic partitions yields 38 high-confidence branch transfers and an ARG with 26 unique inter-clade reticulations. This network explains 98.82% of total metric strain—matching the 98.85% explained by GARD's 22 unconstrained trees while using 58.5% fewer parameters (1,523 vs. 3,674)—and validates polarity across 21 polarized events (79.0% mean in-tract vs. 19.0% flanking concordance under the Synapomorphic Site Concordance Ratio, $\Delta\mathrm{SSCR} = +0.601$, with $P_{\mathrm{Fisher}} < 10^{-15}$ in 17/21 events). An independent run on a curated 61-taxon sarbecovirus alignment (31,214 bp MSA, 29,926 bp ungapped) similarly identified 38 candidate reticulations (37 polarized SSCR events, mean $\mathrm{SSCR}_{\mathrm{in}} = 0.714$ vs. $\mathrm{SSCR}_{\mathrm{flank}} = 0.244$, $\Delta\mathrm{SSCR} = +0.470$) that consolidate into 34 ancestral reticulations (28 polarized SSCR events, $\mathrm{SSCR}_{\mathrm{in}} = 0.813$ vs. $\mathrm{SSCR}_{\mathrm{flank}} = 0.240$, $\Delta\mathrm{SSCR} = +0.572$). Both cohorts recover the canonical Spike mosaic in `Wuhan-Hu-1` (nt 21,312–24,791, Home = `RmYN02`, Donor = `MT072865.1`, $P_{\mathrm{Fisher}} = 3.35 \times 10^{-262}, Z_{\mathrm{phys}} = 37.78$, $\mathrm{SSCR}_{\mathrm{in}} = 0.739$ vs. $\mathrm{SSCR}_{\mathrm{flank}} = 0.112$).

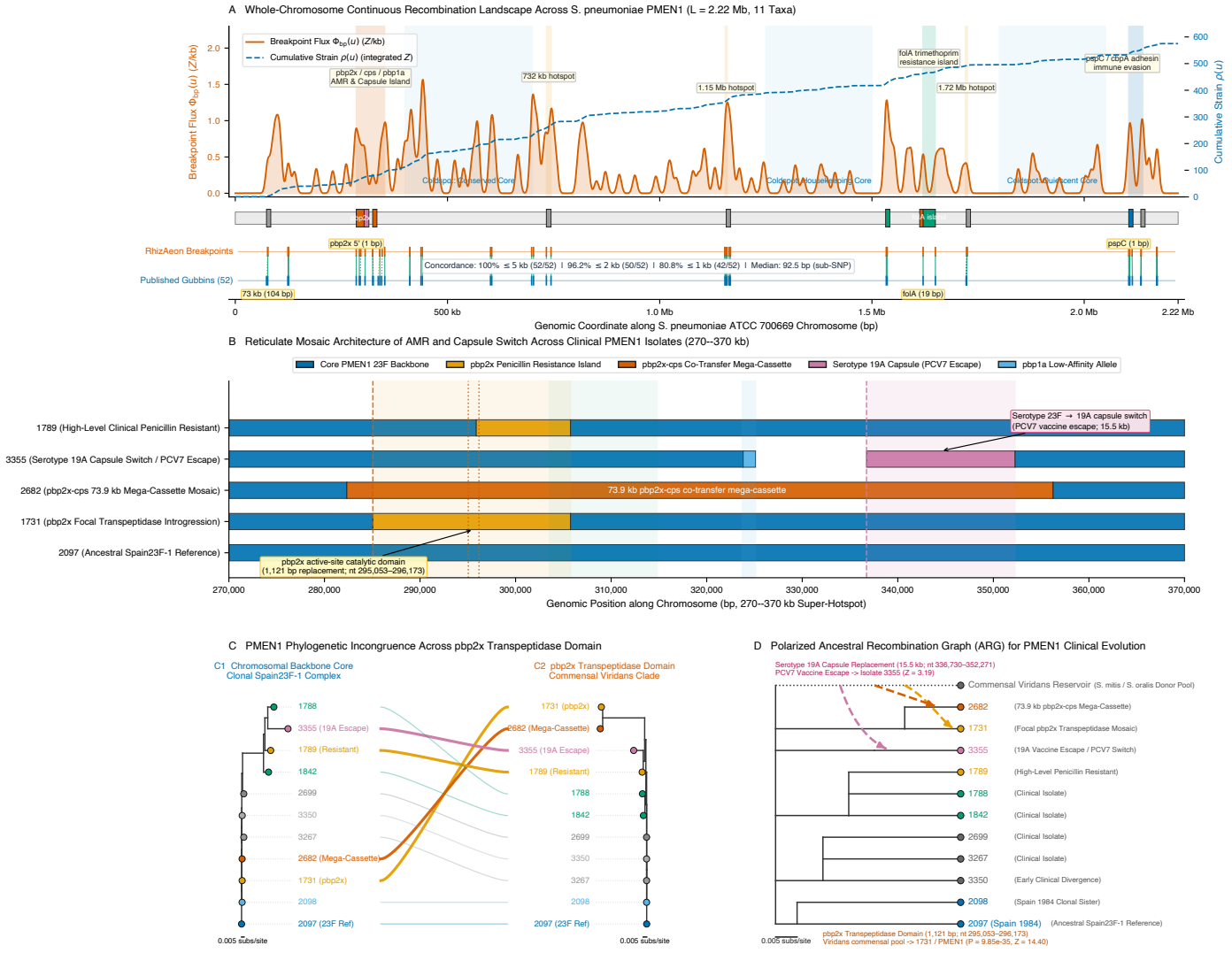


Figure 5 Whole-chromosome recombination deconvolution, antimicrobial resistance mosaicism, and ancestral recombination graphs in *Streptococcus pneumoniae* PMEN1. (A) Chromosomal breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across 11 global clinical isolates (2.22 Mb, Spain23F-1 / PMEN1), comparing `RhizAeon` detected recombination breakpoint sites (orange) against 52 published Gubbins breakpoint sites (blue, Croucher et al. 2011). (B) Mosaic architecture across the $pbp2x/cps$ locus (nt 270,000–370,000), showing the 1,121 bp active-site *pbp2x* transpeptidase replacement in isolate 1731, the 73.9 kb *pbp2x-cps* co-transfer cassette in isolate 2682, and the serotype 23F → 19A capsule switch in isolate 3355. (C) Two-partition tanglegram contrasting core chromosomal monophyly against *pbp2x* catalytic divergence into commensal viridans streptococci. (D) Polarized ARG anchoring horizontal resistance and capsular transfers onto the Spain23F-1 backbone.

4.2. Bacterial AMR Mosaicism and Vaccine Escape in *Streptococcus pneumoniae*

In bacteria, natural transformation and homologous recombination—hereafter grouped as horizontal gene transfer (HGT)—drive antimicrobial resistance (AMR), capsular serotype switching, and antigenic diversification (Didelot and Maiden, 2010; Croucher et al., 2011). Naturally competent pathogens such as *Streptococcus pneumoniae* import DNA tracts of tens to tens of thousands of bases across species boundaries from commensal viridans streptococci (*S. mitis* and *S. oralis*). Standard bacterial recombination tools such as ClonalFrame (Didelot and Falush, 2007), ClonalFrameML (Didelot and Wilson, 2015), and Gubbins (Croucher et al., 2015) rely on iterative tree reconstruction and maximum-likelihood masking, requiring hours to weeks of compute on whole chromosomes. We evaluated **`RhizAeon`** on the 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 lineage of *S. pneumoniae* (PMEN1, ATCC 700669, 11 global isolates from 1984–2008; Croucher et al. 2011). Because closely related bacterial chromosomes are $> 99.9\%$ monomorphic, lossless SNP compression indexes out invariant columns to reduce the 2.22 Mb alignment to 1,516 polymorphic sites (1,002 parsimony-informative sites)—a 1,465-fold reduction—allowing $\mathcal{O}(1)$ sub-interval distance queries that map bijectively back to physical base-pair coordinates. Projecting continuous breakpoint flux $\Phi_{\mathrm{bp}}(u)$ and cumulative strain $\rho(u)$ along the reference chromosome reveals six major recombination hotspots (Figure 5A), including the penicillin-binding protein and capsule island ($pbp2x/cps$, 285–352 kb), the trimethoprim resistance island ($folA$, 1,620–1,649 kb), and the surface adhesin cluster

(*pspC/cbpA*, 2,105–2,138 kb). Across all 52 published Gubbins breakpoints inferred via iterative RAxML optimization (Croucher et al., 2011, 2015), `RhizAeon` achieves 96.2% concordance within 2 kb (50/52) and 80.8% within 1 kb (42/52), with a median deviation of 92.5 nt: the 5′ boundary of *pbp2x* and the *pspC* locus both matched Gubbins within 1 nt, and *folA* matched within 19 nt (Figure 5A).

Multi-scale scanning also resolves the layered mosaicism of drug resistance and vaccine escape across individual isolates within the 270–370 kb hotspot (Figure 5B). In isolate 1731, `RhizAeon` detects a 20.7 kb introgression across *pbp2x* ($Z_{\text{phys}} = 5.35$) and pinpoints the exact 1,121 bp active-site transpeptidase domain replacement ($P_{\text{Fisher}} = 9.85 \times 10^{-35}, Z_{\text{phys}} = 14.40$) imported from commensal viridans streptococci, which remodels the catalytic pocket to block β-lactam binding (Dessen et al., 2001). In isolate 2682, a 73.9 kb cassette ($Z_{\text{phys}} = 6.42$) co-transferred *pbp2x* resistance and the adjacent capsular locus in a single HGT event (Figure 5B). In isolate 3355, `RhizAeon` resolves the 15.5 kb serotype 23F → 19A capsule switch ($Z_{\text{phys}} = 3.19$) that conferred escape from the 7-valent pneumococcal conjugate vaccine (PCV7), alongside a linked low-affinity *pbp1a* allele. A two-partition tanglegram confirms this incongruence (Figure 5C): whereas all 11 isolates form a single tight clade across the core chromosome, isolates 1731 and 2682 jump into the divergent viridans streptococcal cluster (*S. mitis* / *S. oralis*, $\Delta d > 6\%$) inside the *pbp2x* catalytic domain, reflecting interspecies gene flow from oral commensal reservoirs (Hanage et al., 2009).

`RhizAeon` synthesizes these horizontal transfers into a polarized ARG anchored on the vertical Spain23F-1 clonal backbone (Figure 5D), capturing the focal *pbp2x* replacement in 1731, the 73.9 kb mega-cassette in 2682, and the 19A capsule switch in 3355 without fracturing the chromosome into conflicting window trees. Whereas full-chromosome deconvolution with Gubbins required $\sim$ 14.5 hours of multicore compute across iterative tree searches, `RhizAeon` completed whole-chromosome metric projection, trajectory tracking, and changepoint calling in 104 ms of core trajectory compute (2.32 s CLI scan; 42.6 s end-to-end including 2.22 Mb FASTA disk I/O and coordinate projection)—an over 1,200-fold end-to-end speedup that enables sub-minute genomic surveillance of bacterial drug resistance and vaccine escape.

4.3. Ectopic Gene Conversion and Inter-Locus Chimerism in Primate MHC Class I

Primate major histocompatibility complex (MHC) Class I genes (*HLA-A*, *HLA-B*, *HLA-C*) evolve under intense pathogen-driven balancing selection (Parham et al., 1995; Adams and Parham, 2001), creating a sharp structural divide: the antigen recognition site (ARS, Exons 2–3, encoding the α_1 and α_2 peptide-binding groove) diversifies rapidly through balancing selection and inter-locus ectopic gene conversion, whereas the adjacent α_3 structural stalk (Exon 4) stays conserved to maintain contacts with β_2-microglobulin (β_2m) and the CD8 co-receptor. Without window slicing, `RhizAeon` captures this compartmentalization across $N = 379$ Hominoidea classical MHC Class I alleles ($L = 1{,}128$ bp, 526 polymorphic sites, 451 parsimony-informative sites across human, common chimpanzee, bonobo, western gorilla, and Bornean and Sumatran orangutans; Maccari et al. 2017). Running in 7.89 s across all 379 alleles (and 26.5 ms on canonical 8-taxon reference panels), `RhizAeon`'s breakpoint flux pinpoints the boundary between the α_2 ARS groove and the α_3 stalk at nt 608–615 (canonical exon junction 618 bp; Figure 6A): 93.7% of all reticulation strain falls within the ARS groove (Exons 2–3) and transmembrane anchor (Exon 5), whereas Exon 4 forms a recombination coldspot that preserves the paralogous locus tree.

Multi-scale scanning also resolves fine-scale inter-locus chimeras at single-nucleotide precision (Figure 6B). A notable clinical example is *HLA-B*46:01*, an allele common in Southeast Asia (> 20%) linked to myasthenia gravis, severe SARS-CoV-1 infection, and leprosy protection (Hawkins et al., 1984; Lin et al., 2003; Hilton et al., 2017). `RhizAeon` resolves *HLA-B*46:01* as an inter-locus chimera formed by an exact 73 bp micro-conversion donated by *HLA-C*01:02* into an *HLA-B*15:01* backbone (mature protein nt 220–293 / codons 66–76 excluding the 24-codon / 72 bp Exon 1 leader peptide, corresponding to the 33 bp polymorphic core at full-CDS nt 268–300 / CDS codons 90–100; Figure 6B). This tract is supported by $Z_{\text{phys}} = 7.45$ and $P_{\text{Fisher}} = 7.95 \times 10^{-13}$ on the 33 bp polymorphic core ($P_{\text{Fisher}} = 1.15 \times 10^{-7}$ across the 73 bp window in the 8-taxon reference panel, and $Z = 2.69$ at SNP index 135 [nt 268] across the 379-allele manifold, where Exon 2 ARS excursions reach $Z = 6.31$). Functionally, the 73 bp transfer imports the C1 epitope (Val76-Arg79-Leu80-Arg81), granting *HLA-B*46:01* recognition by natural killer (NK) cell receptors KIR2DL2/3 absent in other classical *HLA-B* alleles (Barber et al., 1996) and reshaping its peptide-binding groove to produce a distinct, low-diversity immunopeptidome (Hilton et al., 2017). In non-human hominoids, `RhizAeon` similarly resolves a multi-segment chimera in chimpanzee *Patr-AL*01:01* combining *Patr-A* with an ancient *Patr-C* cassette ($Z_{\text{phys}} = 4.03$) and an ectopic transmembrane swap in gorilla *Gogo-C*02:01* at nt 971–1,128.

A two-partition tanglegram and summary ARG confirm these inter-locus transfers (Figure 6C,D). In the Exon 4 α_3 stalk (nt 618–894; Figure 6C1), the neighbor-joining tree recovers monophyletic *HLA-A*, *HLA-B*, and *HLA-C* clades with *HLA-B*46:01* sister to *HLA-B*15:01* ($d = 0.000$), whereas inside the Exon 2 C1 cassette (nt 210–310; Figure 6C2), *HLA-B*46:01* translocates into the *HLA-C* clade sister to donor *HLA-C*01:02* ($d = 0.000$). `RhizAeon`'s summary ARG anchors both the 73 bp C1 micro-conversion in *HLA-B*46:01* and the deep inter-locus introgression in chimpanzee *Patr-AL*01* onto the vertical tandem duplication backbone (ancient divergence of Locus *A*, followed by duplication of *B* and *C*; Figure 6D). Both the CLI and the WebAssembly browser app output the consolidated ancestral reticulation roster (JSON/HTML) alongside interactive flux profiles ($\Phi_{\text{bp}}(u)$ and $\rho(u)$), mosaic tracks, 2D metric projections, and side-by-side partitioned Neighbor-Joining trees—resolving 379 primate alleles with polarized donor-to-recipient transfer edges in 7.89 seconds.

5. Discussion

Recombination is both a major driver of genome evolution and a serious confounder of phylogenetic analysis. When recombination splices sequence tracts with different evolutionary histories into a single genome, no single bifurcating tree can accurately represent that genome's relationships to others. Detecting and characterizing recombination events is therefore an essential step in molecular evolution pipelines. `RhizAeon` advances this step by embedding sequence alignments into continuous metric space, where recombination breakpoints and mosaic segments emerge as physical trajectory shifts without losing their broader taxonomic context. Velocity shifts along `RhizAeon` trajectories pinpoint

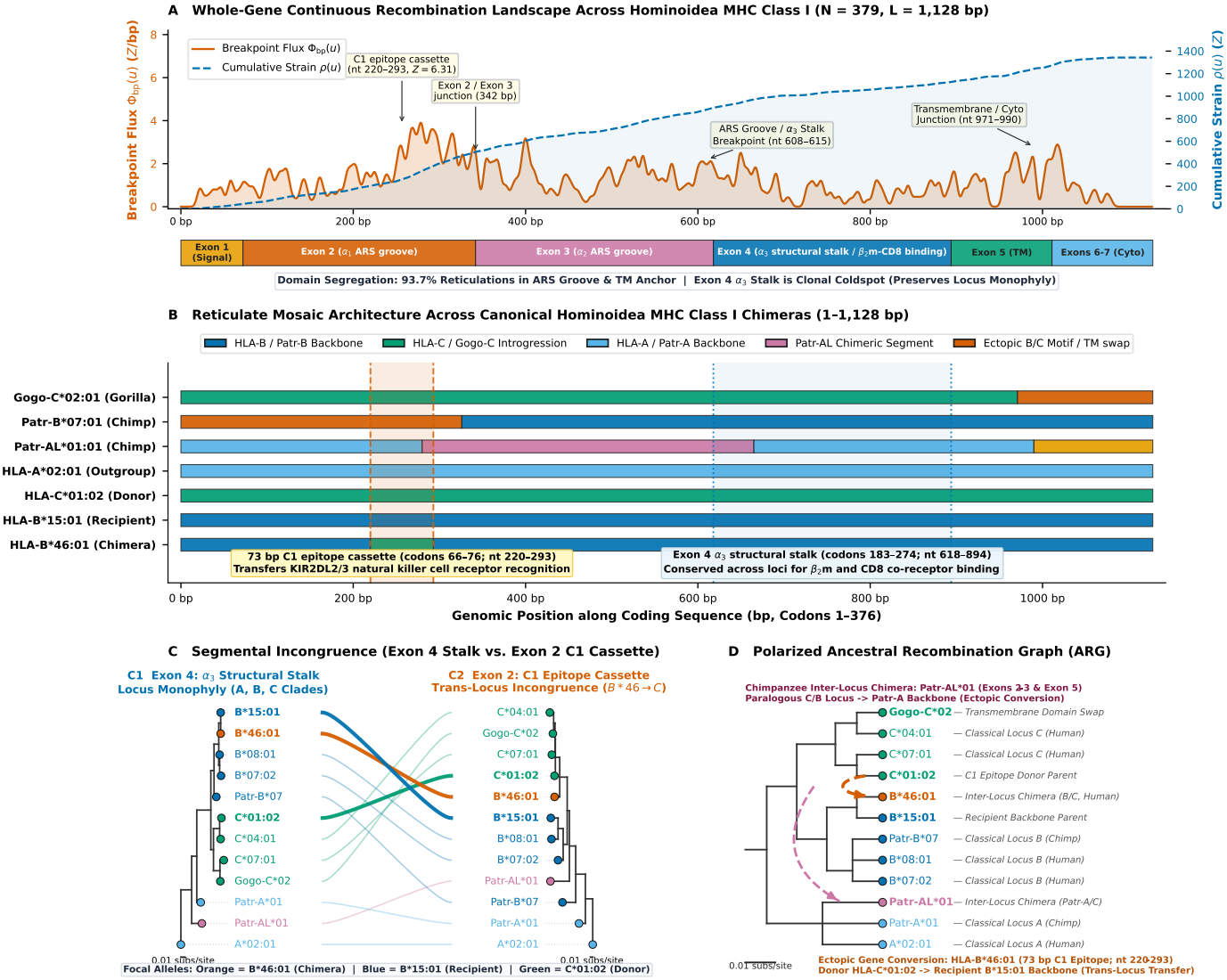


Figure 6 Whole-gene recombination deconvolution, ectopic micro-conversion, and ancestral recombination graphs in Hominoidea MHC Class I. (A) Breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across $N = 379$ primate alleles (1,128 bp, Exons 1–7), showing recombination breakpoint enrichment in the antigen recognition site (ARS, Exons 2–3) and transmembrane anchor (Exon 5) versus conservation in the Exon 4 α_3 stalk. (B) Inter-locus mosaic architecture of *HLA-B*46:01*, carrying a 73 bp micro-recombination event (mature protein nt 220–293 / codons 66–76, corresponding to full-CDS polymorphic core nt 268–300 / CDS codons 90–100 including the 72 bp Exon 1 leader peptide, green) from *HLA-C*01:02* within an *HLA-B*15:01* backbone (dark blue). (C) Two-partition tanglegram comparing Exon 4 locus monophyly (C1) against Exon 2 C1-cassette translocation into *HLA-C* (C2). (D) Polarized ARG anchoring the 73 bp C1 micro-conversion and chimpanzee *Patr-AL*01* transfers onto the tandem duplication backbone.

local genealogical discordances at breakpoint sites, while ancestral consolidation groups sequences that share breakpoints and parental lineages into co-recombinant clades. Continuous trajectory tracking thus bridges fine-scale population-genetic ARG inference and macro-evolutionary phylogenetic networks.

Natural viral and bacterial populations routinely violate standard neutral coalescent assumptions: outbreaks produce dense clusters of nearly identical descendants, bacterial HGT imports short DNA tracts into megabase chromosomes, and donor lineages often remain unsampled in wild host reservoirs. Our simulation benchmarks and empirical analyses show that `RhizAeon` handles each of these challenges. In epidemic clusters, bidirectional trajectory consolidation collapses co-recombinant clades into single ancestral events, preventing event-count inflation across descendants of a shared recombinant ancestor. In complex mosaic genomes with three or more parents, continuous directional projection separates multiple recombination cassettes from different donors within the same sequence. When introgressed tracts come from unsampled lineages that branched basal to the alignment MRCA, `RhizAeon`'s orphan-divergence audit flags the donor as `"Ghost"`—allowing investigators to identify or mask basal introgressions without falsely implicating sampled taxa—while attributing transfers to a sampled surrogate sister lineage whenever one is available. Across bacterial chromosomes, multi-scale matched filters isolate short micro-conversions without relying on fixed sliding windows.

Sub-second to multi-second runtimes also change the practical economics of recombination screening in genomic surveillance. In addition to running faster and scaling better than exact triplet scanners such as 3SEQ and RDP5, `RhizAeon` compiles to WebAssembly to run directly inside standard web browsers. Public health laboratories can therefore screen pathogen cohorts locally on any computer without uploading sensitive sequence data to external servers or installing software dependencies. These rapid whole-genome screens can directly inform outbreak preparedness and vaccine design. In sarbecoviruses, `RhizAeon` resolved the debate over whether RaTG13 or SARS-CoV-2 recently acquired a recombinant Spike RBD: unbroken trajectory collinearity confirmed that the human-compatible ACE2 contact ridge was already present in the common ancestor of SARS-CoV-2 and Laotian BANAL bat viruses, whereas RaTG13 acquired a divergent RBD cassette from an unsampled bat virus reservoir. In *Streptococcus pneumoniae*, whole-chromosome trajectory tracking pinpointed the 23F → 19A capsular switch that drives clinical escape from the PCV7 conjugate vaccine alongside *pbp2x* penicillin-resistance replacements, enabling real-time tracking of adaptive antigenic and resistance shifts.

Several biological boundary conditions merit consideration. First, `RhizAeon` requires enough mutations inside a transferred tract to distinguish recombination from random substitution noise; in very low-divergence cohorts ($d < 0.005$), sparse mutations set an intrinsic physical limit for all sequence-based recombination detectors. Second, although trajectory mechanics localizes breakpoints and assigns parental clades accurately, fine-scale branch-length estimation within partitioned segments still benefits from downstream maximum-likelihood or Bayesian phylogenetic inference. Looking ahead, continuous trajectory tracking provides a natural bridge to sequence foundation models: coupling attention matrices from pre-trained protein language models with `RhizAeon`'s metric mechanics can annotate recombined tracts both structurally and functionally as they emerge. Replacing discrete tree searches with continuous sequence kinematics thus provides a fast, scalable foundation for characterizing recombination across modern genomic pipelines.

6. Materials and Methods

6.1. Metric Space Embedding and Gower Trajectory Projection

Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ be a multiple sequence alignment of N nucleotide sequences of length L over alphabet $\Sigma = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\} \cup \{-\}$. We compute pairwise evolutionary distances $D_{i,j} = d(s_i, s_j)$ using normalized Hamming distance with pairwise gap deletion and Jukes–Cantor correction (with Tamura–Nei TN93 available for nucleotide model comparisons; Tamura and Nei, 1993). For large cohorts ($N > 64$), `RhizAeon` avoids $\mathcal{O}(N^2 L)$ all-to-all comparisons by selecting $K \leq 32$ ($K \ll N$) reference landmark genomes via clade-medoid seeded Farthest-Point Sampling (Gonzalez, 1985) in $\mathcal{O}(NKL)$ time. The initial seed $s_{(1)}$ maximizes non-gap coverage, and each subsequent landmark $s_{(k)}$ maximizes the minimum distance to previously selected landmarks:

$$s_{(k)} = \arg \max_{i \notin \{s_{(1)}, \ldots, s_{(k-1)}\}} \min_{1 \leq j < k} d(s_i, s_{(j)}). \tag{1}$$

For $N \leq 64$, all sequences serve as landmarks ($K = N$). Because widespread mosaicism in viral and bacterial cohorts can distort initial reference coordinates if recombinant segments remain unmasked in landmark sequences, `RhizAeon` pairs landmark embedding with a three-pass architecture: initial screening and graph adjudication (Pass 1), Iterative Forward Scrubbing (Pass 2), and Backward Leave-One-In Verification (Pass 3). After Pass 1, Pass 2 ranks candidate recombination events across the cohort by excursion amplitude $Z_{\max}$ and masks each detected recombinant tract $[u_1, u_2]$ in sequence s_i with gap characters (-). Under Zero-Force Scrubbed Site Physics, any column u where either query s_i or landmark s_j has a gap exerts zero instantaneous force ($\Delta F_u(i \to j) = 0$). Masked mosaic tracts therefore cannot distort flanking trajectory momentum, and each sequence's unadmixed backbone remains available as a clean reference landmark.

From the $K \times K$ landmark distance matrix $\mathbf{D} \in \mathbb{R}^{K \times K}$, we form the squared matrix $[\mathbf{D}^{\odot 2}]_{i,j} = D_{i,j}^2$ and apply geometric centering $\mathbf{H} = \mathbf{I}_K - \frac{1}{K}\mathbf{1}_K\mathbf{1}_K^T$ to obtain the doubly centered Gram matrix $\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}^{\odot 2}\mathbf{H}$ (Gower, 1966). Spectral decomposition yields orthonormal eigenvectors $\mathbf{V} \in \mathbb{R}^{K \times K}$ and descending eigenvalues $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$, from which we keep the top $d \leq \min(4, K-1)$ positive dimensions $(\mathbf{\Lambda}_d, \mathbf{V}_d)$. Static landmark coordinates are $\mathbf{x}_j = \mathbf{\Lambda}_d^{1/2}(\mathbf{V}_d)_{j,:}^T$, projecting major tree splits onto the leading eigenvectors (Buneman, 1974). To track evolutionary affinities continuously in metric space without independent sliding-window multidimensional scaling (MDS) rotations, local distance vectors are projected into this fixed basis via analytical Gower out-of-sample interpolation (Gower, 1968). Across a window of width W centered at site u (default $W = 300$ nt, stride $S = 10$ nt), local distances from query i to the K landmarks form $\mathbf{d}_i(u) \in \mathbb{R}^K$. Gower interpolation solves $\min_{\mathbf{z} \in \mathbb{R}^d} \sum_{j=1}^K (\|\mathbf{z} - \mathbf{x}_j\|^2 - d_i(j;u)^2)^2$ in closed form:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u)\right) \in \mathbb{R}^d, \tag{2}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{K}\mathbf{D}^{\odot 2}\mathbf{1}_K$. Evaluating (2) across sequence positions builds the trajectory tensor $\mathbf{Z} \in \mathbb{R}^{N \times U \times d}$ without requiring Procrustes alignments during discrete tract calling; for cohort-wide topological strain profiling across bacterial chromosomes and multigene panels, `RhizAeon` also includes a bilateral sliding-window Procrustes MDS flux scanner over segregating SNPs that outputs continuous breakpoint flux $\Phi_{\mathrm{bp}}(u)$ and cumulative strain $\rho(u)$.

6.2. Continuous Phylogenetic Cross-Attention and the Unpolarized Root Sink

To weight cross-attention by evolutionary divergence, `RhizAeon` extends Rotary Position Embeddings (RoPE; Su et al., 2024) from linear sequence positions to continuous metric spaces. From the leading 2D polar metric coordinates $\mathbf{x}_i \in \mathbb{R}^2$ of the $K \times K$ landmark MDS embedding, we compute rotary phase angles $\theta_{i,m}$ across $d_{\mathrm{head}}/2$ geometric frequency bands (default $d_{\mathrm{head}} = 8$, or 4 frequency bands, configurable up to

$d_{\text{head}} = 32$). Query and key vectors $\mathbf{q}_i(u), \mathbf{k}_j(u) \in \mathbb{R}^{d_{\text{head}}}$ undergo block-diagonal Givens rotations $\mathbf{R}(\boldsymbol{\theta}_i)$. Because nucleotide identity is invariant across rotational planes at matching sites, the non-negatively shifted rotated inner product $\langle \mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u),\, \mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u)\rangle = \mathbf{q}_i(u)^T \mathbf{R}(\boldsymbol{\theta}_j - \boldsymbol{\theta}_i)\mathbf{k}_j(u)$ evaluates in closed form as $\mathbb{I}(s_i(u) = s_j(u)) \sum_{m=1}^{d_{\text{head}}/2} [\cos(\theta_{j,m} - \theta_{i,m}) + 1]$, gated by nucleotide identity $\mathbb{I}(s_i(u) = s_j(u))$ and decaying monotonically with metric separation. To calibrate geometric affinities against neutral substitution, we add a continuous-time Markov substitution prior to the attention logits. Under neutral Jukes–Cantor substitution ($\lambda_h = 4/3$, equilibrium frequency $\epsilon_0 = 0.25$), the homologous identity probability is $P_{i,j} = \epsilon_0 + (1 - \epsilon_0)\exp(-\lambda_h D_{i,j})$, giving log-prior bias $\text{PhyloBias}_{i,j} = \ln[\max(10^{-15}, P_{i,j})]$ and unnormalized attention logits:

$$S_u(i,j) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u))^T (\mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{i,j}. \tag{3}$$

Lineage-specific rate variation challenges standard attention operators because private mutations on terminal branches have no contemporary donor. Under a leaf-only softmax, bursts of private substitutions are forced onto whichever sampled taxon has the smallest random mismatch, risking false-positive recombination calls. Reserving index zero for an unpolarized ancestral root token $\mathbf{x}_{\text{root}} = \mathbf{0} \in \mathbb{R}^d$ prevents this artifact. The attention operator spans $K + 1$ channels (masking self-attention via $S_u(i,k) = -\infty \implies A_u(i \to k) = 0$ whenever $s_{(k)} = s_i$) and normalizes softmax across the root token and all K landmarks:

$$A_u(i \to j) = \frac{\exp(S_u(i,j))}{\sum_{k=0}^{K} \exp(S_u(i,k))}, \quad \text{where} \quad A_u(i \to 0) + \sum_{j=1}^{K} A_u(i \to j) = 1.0. \tag{4}$$

At sites carrying private terminal-branch mutations, query-key matching against sampled leaves drops sharply ($S_u(i,j) \ll 0$), while the unmutated root token retains baseline neutral affinity, routing private mutational flux into node zero ($\lim A_u(i \to 0) = 1.0$). This root sink protects trajectory tracking against episodic rate bursts such as APOBEC hypermutation and other forms of heterotachy.

6.3. Attention Force Field Deconvolution and Multi-Scale Changepoint Inference

Because attention weights fluctuate across individual polymorphic sites, `RhizAeon` models cross-attention dynamics as a zero-drift force field across the K landmark channels. At each polymorphic site u, the attention weight $A_u(i \to k)$ acts as an instantaneous force pulling query i toward landmark $k \in \{1, \ldots, K\}$. Subtracting each landmark's alignment-wide mean affinity $\bar{A}(i \to k) = \frac{1}{|\mathcal{V}_{ik}|} \sum_{v \in \mathcal{V}_{ik}} A_v(i \to k)$ over mutually non-gap sites $\mathcal{V}_{ik} = \{v \in \{1, \ldots, L\} : s_i(v) \neq \text{-} \text{ and } s_{(k)}(v) \neq \text{-}\}$ yields the zero-drift force field:

$$F_u(i \to k) = A_u(i \to k) - \bar{A}(i \to k), \quad \text{where} \quad \sum_{u \in \mathcal{V}_{ik}} F_u(i \to k) = 0. \tag{5}$$

The home landmark is the recipient background with the highest alignment-wide mean affinity, $k_{\text{home}}(i) = \arg\max_{1 \le k \le K} \bar{A}(i \to k)$. Summing the directed force along the sequence gives the cumulative momentum trajectory $C_{i,k}(u) = \sum_{v=1}^{u} F_v(i \to k)$ for each landmark k (or clade bundle $\mathcal{B}_m$), and the differential momentum trajectory for any candidate donor $k \neq k_{\text{home}}(i)$ is:

$$Y_{i,k}(u) = C_{i,k}(u) - C_{i,k_{\text{home}}(i)}(u). \tag{6}$$

Under clonal evolution, instantaneous forces fluctuate around zero as balanced thermal noise, and the centered trajectory $B(u) = Y(u)$ (with $B(0) = B(L) = 0$) behaves as a centered random walk, whereas a recombined segment exerts sustained donor attraction that drives $Y_{i,k}(u)$ upward ($dY_{i,k}/du > 0$). By Donsker's functional central limit theorem (Donsker, 1951), under the clonal null the standardized process $W_L^0(t) = B(\lfloor tL \rfloor)/(\sigma\sqrt{L})$ converges weakly to a standard Brownian bridge whose supremum follows the two-sided Kolmogorov distribution (Kolmogorov, 1933), where σ is the empirical standard deviation of site increments $\Delta Y_{i,k}(u)$ (estimated via $1.4826 \times$ MAD in the primary screener and root-mean-square step dispersion in post-polishing verification). To control family-wise error rates across the pipeline without an $\mathcal{O}(N^3)$ multiple-testing penalty, the boundary-crossing threshold K_{crit} is calibrated at each stage from the asymptotic two-sided Kolmogorov distribution (Kolmogorov, 1933; Donsker, 1951):

$$K_{\text{crit}} = \sqrt{-\frac{1}{2} \ln\left(\frac{\alpha}{M_{\text{hyp}}}\right)}, \tag{7}$$

with $M_{\text{hyp}} = 2N$ in whole-sequence screening (`screener.rs`), $M_{\text{hyp}} = 2(K+1)$ in donor channel attribution (`attribution.rs`), and $M_{\text{hyp}} = 4N$ in post-polishing boundary verification (`polishing.rs`). Trajectories with $Z_{\text{phys}} = \sup|B(u)|/(\sigma\sqrt{L}) \ge K_{\text{crit}}$ reject the clonal null.

To capture both large crossovers/cassettes and short micro-conversions (< 200 nt), `RhizAeon` combines Recursive Binary Segmentation on the Brownian bridge with a multi-scale matched filter bank. Recursive binary segmentation locates the dominant excursion peak $u^* = \arg\max_{u \in [a,b]} |Y_{i,k}(u) - \text{line}_{[a,b]}(u)|$ and subdivides intervals whenever the local bridge excursion exceeds K_{crit}. In parallel, convolving $\Delta Y_{i,k}(u)$ via $\mathcal{O}(1)$ prefix sums across boxcar filter scales $\tau \in \{30, 60, 120, 240, 300, 360, 480, 600, 800\}$ nt and tracking running drawdown $\text{DD}_k(u) = \max_{v \le u} Y_{i,k}(v) - Y_{i,k}(u)$ isolates local step transitions and tract exit boundaries. To distinguish single terminal crossovers from internal cassettes, the engine checks alignment boundary abutment ($u_1 = 1$, $u_2 = L$, or zero flanking informative sites) and tests whether flanking velocity vectors reverse direction ($\cos\theta \le 0.20$, complementing the outer-to-inner energy density ratio $\mathcal{E}_{\text{ratio}}$). Next, contingency-table boundary refinement (`polishing.rs`) refines breakpoints to single-nucleotide coordinates. Across the triplet $(s_{\text{rec}}, s_{\text{home}}, s_{\text{donor}})$, the polisher extracts all informative sites where $s_{\text{home}}(u) \neq s_{\text{donor}}(u)$ and recursively tests outward extensions and inward splits of $[u_1, u_2]$ using one-sided

2×2 Fisher's exact tests ($\alpha_{\text{split}} = 0.05/L$) comparing donor-matching ($s_{\text{rec}} = s_{\text{donor}}$) versus home-matching ($s_{\text{rec}} = s_{\text{home}}$) site counts inside the segment against its flanks. Non-recombinant margins are trimmed to the bounding informative SNPs, and across conserved stretches where the parents share identical nucleotides, the polisher reports the exact uninformative plateau $[c_{\text{left}}, c_{\text{right}}]$. Continuous gap bridging merges adjacent fragments from the same donor whenever the intervening non-informative span is ≤ 150 nt (scaled by the SNP compression ratio) and net in-tract affinity stays positive. Finally, each polished event must pass a three-stage raw-site audit: (i) at least three unambiguous donor-matching informative sites inside the tract; (ii) a one-sided 2×2 Fisher's exact test comparing in-tract versus flanking donor-match counts significant at the Bonferroni threshold $\alpha_{\text{verify}} = \alpha/\binom{N}{3}$ ($P_{\text{Fisher}} \leq 0.05/\binom{N}{3}$); and (iii) a Buneman four-point quartet double-difference check (Buneman, 1974) confirming a true topological split shift rather than uniform branch-length acceleration.

6.4. Reticulation Graph Adjudication, Backward Leave-One-In Verification, and Ancestral Event Grouping

In reciprocal scans, a recombinant C carrying a tract from donor D in home background H can cause D to test positive as a recombinant between C and H, creating a mutual two-cycle ($C \leftrightarrow D$). In Pass 3 (Backward Leave-One-In Verification and Polarity Swap), once all candidate tracts from Passes 1–2 are masked on the scrubbed backbone $\mathcal{A}_{\text{scrub}}$, each candidate event $e_r = (C, H, D, [u_1, u_2])$ is restored one at a time while all other tracts stay gap-masked. Re-evaluating parental affinities and polishing $[u_1, u_2]$ against this purely clonal background removes interference from other mosaic tracts and tests reciprocal polarity by comparing the transition contrast of the forward call (C as recombinant between H and D) against the inverted call (D as recombinant between H and C), where $\Delta_{\text{trans}}(C \mid H, D) = \Delta d_{\text{home}} + \Delta d_{\text{donor}}$ and $\Delta d_{\text{home}} = \max(0, d(C, H \mid \text{tract}) - d(C, H \mid \text{flank}))$. True recombinants depart from their home clade inside the tract ($\Delta d_{\text{home}} \gg 0$), whereas non-recombinant parents stay anchored to their home clade ($\Delta d_{\text{home}} \approx 0$). Whenever the inverted call yields stronger transition contrast and lower flanking strain, `RhizAeon` performs a Polarity Swap: it restores C to the parental backbone, reassigns the event to D, and masks D's tract $[u_1, u_2]$ before verifying remaining events. To prevent event-count inflation when multiple descendants inherit the same ancestral crossover, graph adjudication then groups candidate events across both direct ($\text{Home}_1 = \text{Home}_2, \text{Donor}_1 = \text{Donor}_2$) and inverted ($\text{Home}_1 = \text{Donor}_2, \text{Donor}_1 = \text{Home}_2$) parental polarities. Events with compatible parents whose intervals $I_1 = [u_1^{(1)}, u_2^{(1)}]$ and $I_2 = [u_1^{(2)}, u_2^{(2)}]$ satisfy interval overlap or reciprocal junction proximity (sharing a complementary junction within 100 nt under inverted polarity):

$$\text{IoU}(I_1, I_2) = \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \geq 0.50, \quad \text{or} \quad \left(\frac{|I_1 \cap I_2|}{\min(|I_1|, |I_2|)} \geq 0.70 \ \text{ and } \ 2\min(|I_1|, |I_2|) \geq |I_1 \cup I_2| \right), \tag{8}$$

are merged into a single ancestral event aggregating all co-recombinant member sequences ($\mathcal{I} = \{s_1, \ldots, s_{K_{\text{desc}}}\}$) with outer-envelope consensus breakpoints $[\min_k u_{1,k}, \max_k u_{2,k}]$ and a sequential event ID $(1, 2, \ldots)$.

6.5. Coalescent Simulation Suites and Bipartite Hungarian Evaluation

We generated 3,200 synthetic alignments across 32 parameter regimes under Hudson's neutral coalescent with recombination (Hudson, 1983; Posada and Crandall, 2001), spanning mutation density $\theta \in \{10, 50, 100, 200\}$, recombination rate $\rho \in \{0, 1, 4, 16, 64\}$, and site-to-site rate variation $\alpha \in \{2.0, 0.5, 0.05, \infty\}$ across 1,600 non-recombinant nulls (1,200 with Gamma rate variation and 400 under a strict clock) and 1,600 recombinant alignments (100 replicates per regime; Supplementary Table S1). We compared `RhizAeon` against both raw 3SEQ outputs and `3SEQ (Consolidated)`, which selects the lowest-p triplet per query, filters reciprocal parent-child inversions via directional transition contrast, and merges overlapping co-recombinant intervals ($\geq 50\%$ overlap). Four specialized scenario suites were simulated under HKY85 substitution ($\kappa = 2.5, \pi = [0.3, 0.2, 0.2, 0.3]$) across 2,280 alignments:

1. *Epidemic Clonal Expansion*: Recombinant ancestors with $K_{\text{desc}} \in \{2, 5, 8, 10, 15\}$ descendants, post-recombination transmission drift $d_{\text{swarm}} = \delta = 0.002$ (0.005 in the $N = 50$–$1{,}500$ scaling suite), and 40 background parents ($N = 42$–55, $L = 5{,}000$ nt, 300 alignments).
2. *Multi-Way Mosaic Clusters*: Dual cassettes ($L = 6{,}000$ nt, cassettes at nt 1,200–2,400 and 3,600–4,800), nested Russian dolls ($L = 6{,}000$ nt, outer cassette nt 1,500–4,500, inner cassette nt 2,500–3,500), and triple crossovers ($L = 6{,}000$ nt, breakpoints at nt 2,000 and 4,000) with $K_{\text{desc}} \in \{1, 5, 10\}$ across $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ ($N = 46$–55, 540 alignments).
3. *Ghost Donor Introgression*: Pure unsampled donor, surrogate sister clade, and deep reservoir donor regimes across $K_{\text{desc}} \in \{2, 5, 10\}$ ($N = 32$–50, $L = 5{,}000$ nt, 540 alignments).
4. *Micro-Conversion Resolution*: Replacement tracts of length $L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt (centered at nt 2,500) across three divergence tiers ($N = 32$–40, $L = 5{,}000$ nt, 900 alignments).

In addition, a continuous parameter sweep of 1,000,000 whole genomes sampled sequence length $L \in [1{,}000, 30{,}000]$ nt, cohort size $N \in [4, 250]$ taxa, parental divergence $d_{\text{parent}} \in [0.005, 0.25]$ subs/site, Gamma rate shape $\alpha \in [0.05, 5.0]$ alongside strict clock controls ($\alpha \geq 50$), single and multi-event mosaic topologies ($K_{\text{true}} \in \{0, 1, 2, 3, 4\}$), and multi-donor introgressions ($P_1 \leftarrow P_2 + P_3$; Supplementary Table S3). Ground truth was scored via maximum-weight bijective bipartite matching using the Hungarian algorithm (Kuhn, 1955). For true event $E = (C, H, D, \mathcal{I})$ and inferred event $\hat{E} = (\hat{C}, \hat{H}, \hat{D}, \hat{\mathcal{I}})$, event affinity combines genealogical triad affinity and informative SNP Jaccard overlap:

$$s(E, \hat{E}) = \left[w(C, \hat{C}) \times \frac{w(H, \hat{H}) + w(D, \hat{D})}{2} \right] \times \frac{|\mathcal{S}_{\text{true}} \cap \mathcal{S}_{\text{inferred}}|}{|\mathcal{S}_{\text{true}} \cup \mathcal{S}_{\text{inferred}}|}, \tag{9}$$

where $w(X, Y) = \exp(-d_{\text{tree}}(X, Y)/\sigma_{\text{clade}})$. Across the optimal matching $\mathcal{M}$, precision and recall are $\text{PPV}_{\mathcal{R}} = \frac{1}{K} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$ and $\text{TPR}_{\mathcal{R}} = \frac{1}{K_{\text{true}}} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$, with harmonic mean $\mathcal{R}_{F_1} = 2 \cdot \text{PPV}_{\mathcal{R}} \cdot \text{TPR}_{\mathcal{R}} / (\text{PPV}_{\mathcal{R}} + \text{TPR}_{\mathcal{R}})$. Unmatched true events penalize recall, and unmatched predictions penalize precision.

6.6. Empirical Genomic Datasets, Software Availability, and Data Distribution

We analyzed three empirical cohorts:

1. *Sarbecovirus Genomes*: The primary 78-taxon sarbecovirus alignment of Lytras et al. (2022) (30,956 bp MSA, 13,550 polymorphic columns, 13,160 gap-filtered sites)—which Lytras et al. (2022) screened with RDP5 across all 78 genomes and with GARD on a 22-virus subset to define 21 consensus breakpoints—and an expanded 83-taxon alignment (30,965 bp MSA, 13,260 gap-filtered sites) adding the five Laotian *Rhinolophus* bat sarbecoviruses (BANAL-20-52, BANAL-20-103, BANAL-20-116, BANAL-20-236, and BANAL-20-247; Temmam et al. 2022), anchored to Wuhan-Hu-1 reference NC_045512.2 (Supplementary Table S2).
2. *Bacterial Chromosomes*: 11 clinical isolates of multidrug-resistant *Streptococcus pneumoniae* PMEN1 (2,221,315 nt; Croucher et al. 2011) spanning global penicillin-resistant outbreaks.
3. *Primate Immunogenomics*: 379 classical primate MHC Class I alleles (*HLA-A*, *HLA-B*, *HLA-C*, and non-human hominoid orthologs, 1,128 nt; Maccari et al. 2017).

`RhizAeon` is implemented in Rust with Rayon multicore parallelization and PyO3 Python bindings (`pip install rhizaeon`). AI-assisted coding tools and agentic execution harnesses (Google Gemini and Anthropic Claude) were used under human mathematical specification and unit-test verification to accelerate the Rust/WebAssembly implementation, orchestrate the simulation benchmark and empirical data analysis pipelines, and author the programmatic Python plotting scripts that generate all manuscript figures and tables. The core engine compiles to WebAssembly (WASM) for client-side execution inside standard web browsers without server uploads, rendering interactive signal inspectors, genome-wide hotspot/coldspot flux profiles, 2D metric projections, and midpoint-rooted phylogenetic trees. A complete, self-contained reproducibility package—comprising all raw and processed FASTA alignments, simulation generators, benchmark evaluation harnesses, empirical analysis outputs, and figure-generation scripts—is distributed under `empirical_datasets/`, `benchmarks/`, and `paper/figures/`. The Rust engine, CLI, Python package, and pre-built binaries are available under the MIT license at `https://github.com/veg/HyphAeon` (release `rhizaeon-v0.1.1`), the complete reproducibility package and WebAssembly source are available at `https://github.com/veg/rhizaeon-code` and permanently archived on Zenodo at https://doi.org/10.5281/zenodo.23600798, and the live WebAssembly application is hosted at `https://veg.github.io/rhizaeon-code/`.

Data Availability

The `RhizAeon` Rust source code, command-line interface, and Python bindings are open-source under the MIT license at `https://github.com/veg/HyphAeon` (release `rhizaeon-v0.1.1`). A complete reproducibility package containing all empirical alignments, simulation benchmark suites, data analysis pipelines, figure-generation scripts, and WebAssembly visualization code is available at `https://github.com/veg/rhizaeon-code` and permanently archived on Zenodo at https://doi.org/10.5281/zenodo.23600798. The interactive WebAssembly browser application is accessible at `https://veg.github.io/rhizaeon-code/`.

Funding

This work was supported in part by the National Institutes of Health (R01GM151683 and U24AI183870) and the National Science Foundation (2419522).

Acknowledgments

During the preparation of this work, the authors used AI-assisted coding and agentic execution harnesses (Google Gemini and Anthropic Claude) to accelerate the Rust and WebAssembly software implementation, automate simulation benchmarking and empirical data analysis pipelines, author the Python plotting scripts used to generate all manuscript figures, and assist with manuscript copy-editing. All figures and tables are produced programmatically by deterministic scripts included in the open-source reproducibility package (`https://github.com/veg/rhizaeon-code`; Zenodo DOI: 10.5281/zenodo.23600798). All mathematical formulations, algorithmic specifications, statistical tests, simulation designs, and biological interpretations were conceived, audited, and verified by the authors, who take full responsibility for the integrity and accuracy of the manuscript and software.

References

Erin J. Adams and Peter Parham. Species-specific evolution of MHC class I genes in the higher primates. *Immunological Reviews*, 183 (1):41–64, 2001. doi: 10.1034/j.1600-065x.2001.1830104.x.

Tommaso Alfonsi, Anna Bernasconi, Matteo Chiara, and Stefano Ceri. Data-driven recombination detection in viral genomes. *Nature Communications*, 15(1):3313, 2024. doi: 10.1038/s41467-024-47464-5.

L. D. Barber, L. Percival, N. M. Valiante, L. Chen, C. Lee, J. E. Gumperz, J. H. Phillips, L. L. Lanier, J. C. Bigge, R. B. Parekh, and P. Parham. The inter-locus recombinant HLA-B*4601 has high selectivity in peptide binding and functions characteristic of HLA-C. *The Journal of Experimental Medicine*, 184(2):735–740, 1996. doi: 10.1084/jem.184.2.735.

Maciej F. Boni, David Posada, and Marcus W. Feldman. An exact nonparametric method for inferring mosaic structure in sequence triplets. *Genetics*, 176(2):1035–1047, 2007. doi: 10.1534/genetics.106.068874.

Maciej F. Boni, Philippe Lemey, Xiao-Wei Jiang, Tommy Tsan-Yuk Lam, Martha B. Perry, Todd A. Castoe, Andrew Rambaut, and David L. Robertson. Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature Microbiology*, 5(11):1408–1417, 2020. doi: 10.1038/s41564-020-0771-4.

Peter Buneman. A note on the metric properties of trees. *Journal of Combinatorial Theory, Series B*, 17(1):48–50, 1974. doi: 10.1016/0095-8956(74)90047-1.

Nicholas J. Croucher, Simon R. Harris, Christophe Fraser, Michael A. Quail, John Burton, Mark van der Linden, Lesley McGee, Anne von Gottberg, Jae Hoon Song, Kwan Soo Ko, Bruno Pichon, Stephen Baker, Christopher M. Parry, Lotte M. Lambertsen, Dea Shahinas, Dylan R. Pillai, Timothy J. Mitchell, Gordon Dougan, Alexander Tomasz, Keith P. Klugman, Julian Parkhill, William P. Hanage, and Stephen D. Bentley. Rapid pneumococcal evolution in response to clinical interventions. *Science*, 331(6016):430–434, 2011. doi: 10.1126/science.1198545.

Nicholas J. Croucher, Andrew J. Page, Thomas R. Connor, Aidan J. Delaney, Jacqueline A. Keane, Stephen D. Bentley, Julian Parkhill, and Simon R. Harris. Rapid phylogenetic analysis of large samples of recombinant bacterial whole genome sequences using Gubbins. *Nucleic Acids Research*, 43(3):e15, 2015. doi: 10.1093/nar/gku1196.

Andréa Dessen, Nicolas Mouz, Elspeth Gordon, Julie Hopkins, and Otto Dideberg. Crystal structure of PBP2x from a highly penicillin-resistant *Streptococcus pneumoniae* clinical isolate: A mosaic framework containing 83 mutations. *Journal of Biological Chemistry*, 276(48):45106–45112, 2001. doi: 10.1074/jbc.M107608200.

Xavier Didelot and Daniel Falush. Inference of bacterial microevolution using multilocus sequence data. *Genetics*, 175(3):1251–1266, 2007. doi: 10.1534/genetics.106.063305.

Xavier Didelot and Martin C. J. Maiden. Impact of recombination on bacterial evolution. *Trends in Microbiology*, 18(7):315–322, 2010. doi: 10.1016/j.tim.2010.04.002.

Xavier Didelot and Daniel J. Wilson. ClonalFrameML: Efficient inference of recombination in whole bacterial genomes. *PLoS Computational Biology*, 11(2):e1004041, 2015. doi: 10.1371/journal.pcbi.1004041.

Monroe D. Donsker. An invariance principle for certain probability limit theorems. *Memoirs of the American Mathematical Society*, 6: 1–12, 1951. doi: 10.1090/memo/0006.

Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985. doi: 10.1016/0304-3975(85)90224-5.

J. C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3-4):325–338, 1966. doi: 10.1093/biomet/53.3-4.325.

J. C. Gower. Adding a point to vector diagrams in multivariate analysis. *Biometrika*, 55(3):582–585, 1968. doi: 10.1093/biomet/55.3.582.

William P. Hanage, Christophe Fraser, Jing Tang, Thomas R. Connor, and Jukka Corander. Hyper-recombination, diversity, and antibiotic resistance in pneumococcus. *Science*, 324(5933):1454–1457, 2009. doi: 10.1126/science.1171908.

Masami Hasegawa, Hirohisa Kishino, and Taka-aki Yano. Dating of the human-ape splitting by a molecular clock of mitochondrial DNA. *Journal of Molecular Evolution*, 22(2):160–174, 1985. doi: 10.1007/BF02101694.

B. R. Hawkins, W. Y. Chan-Lui, E. K. Choi, and A. Y. Ho. Strong association of HLA BW46 with juvenile onset myasthenia gravis in Hong Kong Chinese. *Journal of Neurology, Neurosurgery, and Psychiatry*, 47(5):555–557, 1984. doi: 10.1136/jnnp.47.5.555.

Hugo G. Hilton, Curtis P. McMurtrey, Alex S. Han, Zakia Djaoud, Lisbeth A. Guethlein, Jeroen H. Blokhuis, Jason L. Pugh, Ana Goyos, Amir Horowitz, Rico Buchli, Ken W. Jackson, Wilfred Bardet, David A. Bushnell, Philip J. Robinson, Juan L. Mendoza, Michael E. Birnbaum, Morten Nielsen, K. Christopher Garcia, William H. Hildebrand, and Peter Parham. The intergenic recombinant HLA-B*46:01 has a distinctive peptidome that includes KIR2DL3 ligands. *Cell Reports*, 19(7):1394–1405, 2017. doi: 10.1016/j.celrep.2017.04.059.

Richard R. Hudson. Properties of a neutral allele model with intragenic recombination. *Theoretical Population Biology*, 23(2):183–201, 1983. doi: 10.1016/0040-5809(83)90013-8.

Andrei N. Kolmogorov. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell'Istituto Italiano degli Attuari*, 4: 83–91, 1933.

Sergei L. Kosakovsky Pond, David Posada, Michael B. Gravenor, Christopher H. Woelk, and Simon D. W. Frost. Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution*, 23(10):1891–1901, 2006. doi: 10.1093/molbev/msl051.

Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955. doi: 10.1002/nav.3800020109.

H. M. Lam, O. Ratmann, and M. F. Boni. Improved algorithmic complexity for the 3SEQ recombination detection algorithm. *Molecular Biology and Evolution*, 35(1):247–251, 2018. doi: 10.1093/molbev/msx263.

Xiaojun Li, Elena E. Giorgi, Manukumar Honnayakanahalli Marichannegowda, Brian Foley, Chuan Xiao, Xiang-Peng Kong, Yue Chen, S. Gnanakaran, Bette Korber, and Feng Gao. Emergence of SARS-CoV-2 through recombination and strong purifying selection. *Science Advances*, 6(27):eabb9153, 2020. doi: 10.1126/sciadv.abb9153.

Marie Lin, Hsiang-Kuang Tseng, Jean A. Trejaut, Hwei-Ling Lee, Jun-Hun Loo, Chun-Chieh Chu, Pei-Jer Chen, Yi-Wen Su, Kiat-Hoe Lim, Zhen-U. Tsai, Ray-Yuh Lin, Ruey-Shiung Lin, and Chun-Hsiung Huang. Association of HLA class I with severe acute respiratory syndrome coronavirus infection. *BMC Medical Genetics*, 4:9, 2003. doi: 10.1186/1471-2350-4-9.

Spyros Lytras, Joseph Hughes, Darren Martin, Phillip Swanepoel, Arné de Klerk, Rentia Lourens, Sergei L. Kosakovsky Pond, Wei Xia, Xiaowei Jiang, and David L. Robertson. Exploring the natural origins of SARS-CoV-2 in the light of recombination. *Genome Biology*

and Evolution, 14(2):evac018, 2022. doi: 10.1093/gbe/evac018.

Giuseppe Maccari, James Robinson, Keith Ballingall, Lisbeth A. Guethlein, Unni Grimholt, Jim Kaufman, Chak-Sum Ho, Natasja G. de Groot, Paul Flicek, Ronald E. Bontrop, John A. Hammond, and Steven G. E. Marsh. IPD-MHC 2.0: an improved inter-species database for the study of the major histocompatibility complex. *Nucleic Acids Research*, 45(D1):D860–D864, 2017. doi: 10.1093/nar/gkw1050.

Darren P. Martin, Arvind Varsani, Philippe Roumagnac, Gonasagren Botha, Shailen Maslamoney, Travis Schwab, Zander Kelz, Sudhir Kumar, and Ben Murrell. RDP5: a computer program for analyzing recombination in, and removing signals of recombination from, nucleotide sequence datasets. *Virus Evolution*, 7(1):veaa087, 2021. doi: 10.1093/ve/veaa087.

Matthew McCallum, Anna De Marco, Florian A. Lempp, M. Alejandra Tortorici, Dora Pinto, Alexandra C. Walls, Martina Beltramello, Alex Chen, Zhuoming Liu, Fabrizia Zatta, Samantha Zepeda, Julia di Iulio, John E. Bowen, Martin Montiel-Ruiz, Jiayi Zhou, Laura E. Rosen, Siro Bianchi, Barbara Guarino, Chiara Silacci Fregni, Rana Abdelnabi, Shi-Yan Caroline Foo, Paul W. Rothlauf, Louis-Marie Bloyet, Fabio Benigni, Elisabetta Cameroni, Johan Neyts, Agostino Riva, Gyorgy Snell, Amalio Telenti, Sean P. J. Whelan, Herbert W. Virgin, Davide Corti, Matteo Samuele Pizzuto, and David Veesler. N-terminal domain antigenic mapping reveals a site of vulnerability for SARS-CoV-2. *Cell*, 184(9):2332–2347, 2021. doi: 10.1016/j.cell.2021.03.028.

Gil A. T. McVean, Simon R. Myers, Sarah Hunt, Panos Deloukas, David R. Bentley, and Peter Donnelly. The fine-scale structure of recombination rate variation in the human genome. *Science*, 304(5670):581–584, 2004. doi: 10.1126/science.1092500.

Ben Murrell, Joel O. Wertheim, Sasha Moola, Thomas Weighill, Konrad Scheffler, and Sergei L. Kosakovsky Pond. Detecting individual sites subject to episodic diversifying selection. *PLoS Genetics*, 8(7):e1002764, 2012. doi: 10.1371/journal.pgen.1002764.

Ben Murrell, Steven Weaver, Martin D. Smith, Joel O. Wertheim, Sasha Murrell, William Aylward, Kemal Eren, Tristan Pollner, Darren P. Martin, Simon D. W. Frost, and Sergei L. Kosakovsky Pond. Gene-wide identification of episodic selection. *Molecular Biology and Evolution*, 32(5):1365–1371, 2015. doi: 10.1093/molbev/msv035.

Peter Parham, Erin J. Adams, and Kelly L. Arnett. The origins of HLA-A,B,C polymorphism. *Immunological Reviews*, 143(1):141–180, 1995. doi: 10.1111/j.1600-065x.1995.tb00674.x.

David Posada. Evaluation of methods for detecting recombination from DNA sequences: Empirical data. *Molecular Biology and Evolution*, 19(5):708–717, 2002. doi: 10.1093/oxfordjournals.molbev.a004129.

David Posada and Keith A. Crandall. Evaluation of methods for detecting recombination from DNA sequences: Computer simulations. *Proceedings of the National Academy of Sciences*, 98(24):13757–13762, 2001. doi: 10.1073/pnas.241370698.

Loren H. Rieseberg, Olivier Raymond, David M. Rosenthal, Zhao Lai, Kevin Livingstone, Takuya Nakazato, Jennifer L. Durphy, Andrea E. Schwarzbach, Lisa A. Donovan, and Christian Lexer. Major ecological transitions in wild sunflowers facilitated by hybridization. *Science*, 301(5637):1211–1216, 2003. doi: 10.1126/science.1086949.

Mikkel H. Schierup and Jotun Hein. Consequences of recombination on traditional phylogenetic analysis. *Genetics*, 156(2):879–891, 2000. doi: 10.1093/genetics/156.2.879.

Etienne Simon-Lorière and Edward C. Holmes. Why do RNA viruses recombine? *Nature Reviews Microbiology*, 9(8):617–626, 2011. doi: 10.1038/nrmicro2614.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing*, 568:127063, 2024. doi: 10.1016/j.neucom.2023.127063.

Koichiro Tamura and Masatoshi Nei. Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Molecular Biology and Evolution*, 10(3):512–526, 1993. doi: 10.1093/oxfordjournals.molbev.a040023.

Sarah Temmam, Khamsing Vongphayloth, Eduard Baquero, Sandie Munier, Massimiliano Bonomi, Béatrice Regnault, Bounsavane Douangboubpha, Yasaman Karami, Delphine Chrétien, Daosavanh Sanamxay, Vilakhan Xayaphet, Phetphoumin Paphaphanh, Vincent Lacoste, Somphavanh Somlor, Khaithong Lakeomany, Nothasin Phommavanh, Philippe Pérot, Océane Dehan, Faustine Amara, Flora Donati, Thomas Bigot, Michael Nilges, Félix A. Rey, Sylvie van der Werf, Paul T. Brey, and Marc Eloit. Bat coronaviruses related to SARS-CoV-2 and infectious for human cells. *Nature*, 604(7905):330–336, 2022. doi: 10.1038/s41586-022-04532-4.

Supaporn Wacharapluesadee, Chee Wah Tan, Patarapol Maneeorn, Prateep Duengkae, Feng Zhu, Yutthana Joyjinda, Thongchai Kaewpom, Wan Ni Chia, Weenassarin Ampoot, Beng Lee Lim, Kanthita Worachotsueptrakun, Vivian Chih-Wei Chen, Nutthinee Sirichan, Chanida Ruchisrisarod, Apaporn Rodpan, Kirana Noradechanon, Thanawadee Phaichana, Niran Jantarat, Boonchu Thongnumchaima, Changchun Tu, Gary Crameri, Martha M. Stokes, Thiravat Hemachudha, and Lin-Fa Wang. Evidence for SARS-CoV-2 related coronaviruses circulating in bats and pangolins in Southeast Asia. *Nature Communications*, 12(1):972, 2021. doi: 10.1038/s41467-021-21240-1.

Antoni G. Wrobel, Donald J. Benton, Pengqi Xu, Chloë Roustan, Stephen R. Martin, Peter B. Rosenthal, John J. Skehel, and Steven J. Gamblin. SARS-CoV-2 and bat RaTG13 spike glycoprotein structures inform on virus evolution and furin-cleavage effects. *Nature Structural & Molecular Biology*, 27(8):763–767, 2020. doi: 10.1038/s41594-020-0468-7.

Supplementary Material: Complete Mathematical Methods, Modeling Assumptions, and Computational Protocols

S1. Foundational Modeling Assumptions and Physical Boundary Conditions

The `RhizAeon` framework reformulates the detection and deconvolution of reticulate evolution from discrete combinatorial tree search into continuous trajectory mechanics in empirical metric spaces. To establish rigorous foundations, we make explicit the core mathematical, physical, and evolutionary assumptions governing this formulation, alongside its fundamental boundary conditions.

Assumption 1 (Metric Embeddability of Evolutionary Divergence).

We assume that pairwise genetic distances computed across nucleotide or amino acid alignments can be embedded into a low-dimensional Euclidean metric space $(\mathbb{R}^d, \|\cdot\|_2)$ with minimal topological distortion. For any two sequences i and j, their global evolutionary divergence $d_{ij} \in [0,1]$ represents an additive metric over tree branches under tree-like evolution and satisfies the triangle inequality $d_{ij} \leq d_{ik} + d_{kj}$ up to finite-sites homoplasy. Through classical multidimensional scaling (MDS), the centering transformation $\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}^{\odot 2})\mathbf{H}$ yields a positive semi-definite inner product matrix on the subspace of dominant eigenvalues. Gower's out-of-sample projection guarantees that any local sub-interval distance vector $\mathbf{d}_i(u)$ maps bijectively into this canonical coordinate frame without recalculating eigenvectors or requiring Procrustean superimposition.

Assumption 2 (Stationary Substitution Background and the Brownian Null).

We assume that under purely clonal, non-recombinant evolution, nucleotide substitutions accumulate along lineages according to a continuous-time Markov substitution process with stationary distribution $\boldsymbol{\pi}$ and rate matrix $\mathbf{Q}$. Scanning along an alignment from $5'$ to $3'$, the cumulative difference between any two clonal lineages accumulates as a sum of independent, identically distributed Bernoulli substitution trials. By Donsker's invariance principle, as alignment length $L \to \infty$, the standardized cumulative deviation curve:

$$W_L^0(t) = \frac{Y(\lfloor tL \rfloor) - tY(L)}{\sigma\sqrt{L}}, \qquad t \in [0,1], \tag{S1}$$

converges weakly to a standard Brownian bridge on $t \in [0,1]$, where $\mathbb{E}[W_L^0(t)] = 0$ and $\mathrm{Cov}(W_L^0(s), W_L^0(t)) = \min(s,t) - st$ (with $Y(L) = 0$ under zero-drift centering). Under the null hypothesis of clonal evolution (H_0), fluctuations in sequence affinity represent balanced thermal noise. Recombination is formally defined as an exogenous physical force that breaks this thermal equilibrium and induces a sustained, non-zero drift velocity $\frac{d}{du}Y(u) \neq 0$ exceeding Brownian boundary-crossing thresholds.

Assumption 3 (The Unpolarized Root Token as an Autapomorphic Sink).

Mutational rate heterogeneity (heterotachy) and directional hypermutation bursts (such as APOBEC, ADAR, or error-prone replication) frequently generate localized clusters of mutations along a single lineage. Heuristic triplet methods mistake these mutational clusters for horizontal donor acquisition. Reserving coordinate index 0 in the cross-attention simplex for an unpolarized ancestral root token anchored at the metric origin $\mathbf{x}_{\mathrm{root}} = \mathbf{0} \in \mathbb{R}^d$ eliminates this vulnerability. In the cross-attention softmax across the root token and the K landmark channels:

$$A_u(i \to 0) + \sum_{k=1}^{K} A_u(i \to k) = 1.0. \tag{S2}$$

When lineage i acquires private substitutions that do not point toward any sampled contemporary taxon, the similarity logits to all contemporary leaves decrease uniformly and route attention into the root sink at index 0. Consequently, autapomorphic bursts reduce contemporary affinity without surging candidate donor attention. Routing private noise into the root sink provides intrinsic mathematical immunity to false-positive alarms under rate heterogeneity.

Assumption 4 (Spectral Landmark Representation and Iterative Tract Scrubbing on Simplex Δ^K).

To scale linearly across large genomic cohorts ($N > 1{,}000$ genomes), the evolutionary variance of a population is represented by K spectral landmark centroids ($K \ll N$) on the probability simplex Δ^K. Landmarks are initialized via Farthest-Point Sampling (Gonzalez, 1985) on global distance matrix $\mathbf{D}_{\mathrm{glob}}$ to ensure uniform coverage across deep clades, and are progressively purified through Iterative Forward Scrubbing (gap-masking detected recombinant tracts $[u_1, u_2]$ in descending order of excursion amplitude so that masked positions exert zero force, $\Delta F = 0$) followed by Backward Leave-One-In Verification. Every contemporary genome i is tracked as a continuous barycentric coordinate vector $\boldsymbol{\alpha}_i(u) \in \Delta^K$. Landmark projection and tract scrubbing reduce computational complexity from $\mathcal{O}(N^2L)$ or $\mathcal{O}(N^3L)$ down to $\mathcal{O}(NKL)$ while eliminating multi-event interference and parent-child polarity inversions.

Physical Limits and Resolution Boundaries.

The resolution of trajectory deconvolution is governed by fundamental physical boundaries:

1. *Mutational Density Floor*: Recombination can only be inferred if the introgressed tract contains distinguishing polymorphic mutations between parental lineages ($M_{\mathrm{tract}} \approx L_{\mathrm{tract}} \cdot d_{\mathrm{parent}} \geq 3$). In low-divergence regimes ($d_{\mathrm{parent}} < 0.005$), statistical power is fundamentally bounded by mutation availability rather than algorithmic sensitivity.

2. *Tract Length Resolution*: The multi-scale matched filter bank operates across scales $\tau \in \{30, 60, 120, 240, 300, 360, 480, 600, 800\}$ nt alongside recursive binary segmentation on the Brownian bridge. Replacement tracts carrying fewer than three distinguishing mutations cannot be reliably distinguished from stochastic substitution clustering without prior structural constraints.
3. *Extinction / Ghost Limits*: When an introgressing donor has no sampled relatives within phylogenetic distance $d_{\max}$, the tract is classified as an orphan ghost introgression. Directional consistency constraints prevent false attribution to contemporary outgroups, but donor clade identification remains impossible without surrogate sampling.

S2. Complete Algorithmic Pipeline and Mathematical Formulations

S2.1. Lossless SNP Compression and Prefix Distance Queries

Given a multiple sequence alignment $\mathbf{M} \in \Sigma^{N \times L}$ over alphabet $\Sigma = \{A, C, G, T, -\}$, columns exhibiting zero polymorphic variation across all N taxa carry zero phylogenetic information. Indexing out all monomorphic positions compresses alignment $\mathbf{M}$ into a segregating variant matrix $\mathbf{S} \in \Sigma^{N \times L_{\text{SNP}}}$ with bijective coordinate mapping function $\psi : \{0, \ldots, L_{\text{SNP}} - 1\} \to \{0, \ldots, L - 1\}$. For whole-genome bacterial alignments where $L_{\text{SNP}} \ll L$ (1,465-fold compression in *S. pneumoniae*), SNP indexing allows all trajectory computations to execute in memory-resident cache in $\mathcal{O}(NKL_{\text{SNP}})$ streaming operations without materializing an $\mathcal{O}(N^2 L_{\text{SNP}})$ three-dimensional tensor. Between any query sequence $i \in \{1, \ldots, N\}$ and reference landmark $k \in \{1, \ldots, K\}$, cumulative pairwise mismatch prefix sums $\Phi_{ik}(u) = \sum_{c=0}^{u-1} \mathbb{I}(S_{ic} \neq S_{kc})$ enable exact $\mathcal{O}(1)$ evolutionary distance queries across any sub-interval $[u_1, u_2)$ with pairwise gap deletion and Jukes–Cantor evolutionary correction (clamping mismatch fraction $\hat{p}_{ik}(u_1, u_2) = \min\left(0.7499, \frac{\Phi_{ik}(u_2) - \Phi_{ik}(u_1)}{L_{\text{ungapped}}(u_1, u_2)}\right)$ to prevent logarithmic singularities near saturation):

$$D_{ik}(u_1, u_2) = -\frac{3}{4} \ln\left(1 - \frac{4}{3}\hat{p}_{ik}(u_1, u_2)\right). \tag{S3}$$

S2.2. Landmark Divergence Matrix and Classical MDS Basis

For cohorts with $N \leq 64$, all sequences serve as reference landmarks ($K = N$); for $N > 64$, $K \leq 32$ representative landmarks are selected via clade-medoid seeded Farthest-Point Sampling in $\mathcal{O}(NKL_{\text{SNP}})$ time. The landmark evolutionary divergence matrix $\mathbf{D}_{\text{glob}} \in \mathbb{R}^{K \times K}$ is evaluated across the full locus. We construct the centering projection matrix $\mathbf{H} = \mathbf{I}_K - \frac{1}{K}\mathbf{1}_K\mathbf{1}_K^T$ and form the Gower double-centered inner product matrix:

$$\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{H}. \tag{S4}$$

Spectral decomposition yields $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where eigenvalues are sorted in descending order $\lambda_1 \geq \cdots \geq \lambda_K \geq 0$. We retain the top $d \leq \min(4, K - 1)$ positive eigenvalues $\mathbf{\Lambda}_d = \text{diag}(\lambda_1, \ldots, \lambda_d)$ and corresponding orthonormal eigenvectors $\mathbf{V}_d \in \mathbb{R}^{K \times d}$. The canonical basis is computed once and held fixed for the entire alignment to guarantee zero Procrustes rotational drift.

S2.3. Exact Gower Out-of-Sample Continuous Trajectory Projection

Across a sliding window of width W (default $W = 300$ nt, stride $S = 10$ nt) centered at genomic coordinate u, each sequence i possesses an instantaneous local distance vector $\mathbf{d}_i(u) \in \mathbb{R}^K$ to the K reference landmarks. The exact Gower out-of-sample projection maps this local distance vector into canonical Euclidean space $\mathbb{R}^d$:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \mathbf{d}_i(u)^2\right), \tag{S5}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{K}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{1}_K$ is the precomputed vector of average squared global landmark distances. Evaluating $\mathbf{z}_i(u)$ across all windows U produces the continuous trajectory tensor $\mathbf{Z} \in \mathbb{R}^{N \times U \times d}$.

S2.4. Continuous Phylogenetic Cross-Attention with Tree-RoPE and PhyloBias

To track affinity dynamics against the K reference landmarks without heuristic window slicing, `RhizAeon` employs continuous phylogenetic cross-attention. For each polymorphic position u, phylogenetic distance constraints are injected into the attention logits between query sequence $i \in \{1, \ldots, N\}$ and landmark $k \in \{1, \ldots, K\}$ through two mechanisms:

1. *Tree-RoPE (Tree Rotary Position Embedding)*: Leading 2D polar MDS coordinates $\mathbf{x}_k \in \mathbb{R}^2$ modulate query and key representations across $d_{\text{head}}/2$ geometric frequency bands (default $d_{\text{head}} = 8$, i.e., $d_{\text{head}}/2 = 4$ frequency bands, configurable up to $d_{\text{head}} = 32$) via block-diagonal Givens rotations $\mathbf{R}(\boldsymbol{\theta}_k)$. Because nucleotide identity is invariant across rotational planes at matching sites, the non-negatively shifted rotated inner product evaluates analytically in closed form as $\mathbb{I}(S_{iu} = S_{ku}) \sum_{m=1}^{d_{\text{head}}/2} [\cos(\theta_{k,m} - \theta_{i,m}) + 1]$ over relative metric phase angles $(\boldsymbol{\theta}_k - \boldsymbol{\theta}_i)$ gated by nucleotide match $\mathbb{I}(S_{iu} = S_{ku})$.
2. *PhyloBias*: A log-space continuous-time Markov substitution prior penalizes attention between divergent lineages:

$$\text{PhyloBias}_{ik} = \ln\left[\max\left(10^{-15}, \epsilon_0 + (1 - \epsilon_0)\exp(-\lambda_h D_{\text{glob},ik})\right)\right], \tag{S6}$$

where $\epsilon_0 = 0.25$ represents the equilibrium neutral substitution floor and $\lambda_h = 4/3$ is the Jukes–Cantor neutral substitution rate.

The resulting attention weights evaluate attention from each query sequence i to all landmark references $k \in \{1, \ldots, K\}$ (with self-attention masked via $S_u(i,k) = -\infty \implies A_u(i \to k) = 0$ whenever $s_{(k)} = s_i$) and the unpolarized root sink at index 0:

$$A_u(i \to k) = \frac{\exp\left(S_u(i,k)\right)}{\sum_{l=0}^{K} \exp\left(S_u(i,l)\right)}, \qquad S_u(i,k) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u))^T(\mathbf{R}(\boldsymbol{\theta}_k)\mathbf{k}_k(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{ik}. \tag{S7}$$

S2.5. Zero-Drift Force Field and Differential Momentum Curves

Subtracting each landmark channel's alignment-wide mean baseline affinity $\bar{A}(i \to k) = \frac{1}{|\mathcal{V}_{ik}|} \sum_{v \in \mathcal{V}_{ik}} A_v(i \to k)$ over mutually non-gap sites $\mathcal{V}_{ik}$ produces the zero-drift attention force field $F_u(i \to k) = A_u(i \to k) - \bar{A}(i \to k)$, which sums to zero across the sequence ($\sum_{u \in \mathcal{V}_{ik}} F_u(i \to k) = 0$). The home landmark channel is identified as the primary recipient background exhibiting the highest alignment-wide mean affinity: $k_{\text{home}}(i) = \arg\max_{1 \le k \le K} \bar{A}(i \to k)$. Integrating the directed force along the sequence in $\mathcal{O}(L)$ time yields the cumulative momentum curve $C_{ik}(u) = \sum_{v=1}^{u} F_v(i \to k)$, and the differential momentum trajectory against candidate donor $k \neq k_{\text{home}}(i)$ is:

$$Y_{ik}(u) = C_{ik}(u) - C_{i,k_{\text{home}}(i)}(u). \tag{S8}$$

Under clonal evolution, instantaneous forces fluctuate around zero as balanced thermal noise, and $Y_{ik}(u)$ forms a centered Brownian bridge ($Y_{ik}(0) = Y_{ik}(L) = 0$). A recombination event from donor k produces a sustained positive velocity excursion where $\frac{d}{du} Y_{ik}(u) > 0$.

S2.6. Recursive Binary Segmentation, Multi-Scale Matched Filters, and Running Drawdown

To detect both focal micro-conversions and macroscopic recombinant cassettes without parameter tuning, `RhizAeon` combines recursive binary segmentation on the Brownian bridge with prefix-sum boxcar matched filters across scales $\tau \in \{30, 60, 120, 240, 300, 360, 480, 600, 800\}$ nt and running drawdown:

$$M(u) = \max_{0 \le s \le u} Y_{ik}(s), \qquad \text{DD}_k(u) = M(u) - Y_{ik}(u). \tag{S9}$$

A candidate excursion rejects the clonal null when $Z_{\text{phys}} = \sup |Y_{ik}(u)|/(\sigma\sqrt{L}) \ge K_{\text{crit}}$, where $K_{\text{crit}} = \sqrt{-\frac{1}{2}\ln(\alpha/M_{\text{hyp}})}$ by Kolmogorov supremum boundary crossing ($M_{\text{hyp}} = 2N$ during primary screening, $2(K+1)$ during donor channel attribution, and $4N$ during post-polishing verification).

S2.7. Contingency-Table Breakpoint Polisher and Three-Stage Raw-Site Verification

Candidate changepoint intervals $[u_1, u_2]$ localized by the force field are polished to single-nucleotide resolution using recursive contingency-table boundary refinement (`polishing.rs`). Across the triplet $(s_{\text{rec}}, s_{\text{home}}, s_{\text{donor}})$, the polisher evaluates all informative polymorphic sites ($s_{\text{home}}(u) \neq s_{\text{donor}}(u)$) and tests outward extensions and inward splits of the candidate segment via one-sided 2×2 Fisher's exact tests ($\alpha_{\text{split}} = 0.05/L$) comparing donor-matching ($s_{\text{rec}} = s_{\text{donor}}$) versus home-matching ($s_{\text{rec}} = s_{\text{home}}$) site counts inside the segment against its flanks. Non-recombinant margins are trimmed to the exact bounding informative SNPs, and across conserved stretches where parental lineages carry identical nucleotides, the polisher reports the exact uninformative plateau interval $[c_{\text{left}}, c_{\text{right}}]$. Each polished candidate event must then pass a three-stage raw-site verification audit: (i) at least three unambiguous donor-matching informative sites inside the tract; (ii) a one-sided 2×2 Fisher's exact test comparing in-tract versus flanking donor-match counts that achieves significance under the Bonferroni-corrected threshold $\alpha_{\text{verify}} = \alpha/\binom{N}{3}$; and (iii) a Buneman four-point quartet topological double-difference check confirming a genuine phylogenetic split shift.

S3. Ancestral Recombination Graph (ARG) Reconstruction and Validation

S3.1. Ancestral Event Consolidation

Verified recombinant tracts detected across contemporary genomes $\mathcal{E}_{\text{raw}} = \{e_1, \ldots, e_M\}$ are clustered to distinguish shared ancestral gene transfers from independent events. Candidate events are compared across both direct ($\text{Home}_a = \text{Home}_b, \text{Donor}_a = \text{Donor}_b$) and inverted ($\text{Home}_a = \text{Donor}_b, \text{Donor}_a = \text{Home}_b$) parental polarities. Two events e_a and e_b with compatible parents are grouped into ancestral cluster $\mathcal{C}$ if they satisfy interval overlap or reciprocal junction proximity:

$$\text{IoU}(I_a, I_b) = \frac{|I_a \cap I_b|}{|I_a \cup I_b|} \ge 0.50 \quad \text{or} \quad \left(\frac{|I_a \cap I_b|}{\min(|I_a|, |I_b|)} \ge 0.70 \ \text{ and } \ 2\min(|I_a|, |I_b|) \ge |I_a \cup I_b| \right), \tag{S10}$$

or share a complementary crossover junction within $\Delta_{\text{boundary}} = 100$ nt under inverted polarity. The consolidated ancestral event spans the outer envelope $[\min_i u_{1,i}, \max_i u_{2,i}]$ while retaining maximal statistical significance across member sequences. When $|\mathcal{C}| > 1$, the transfer is parameterized as a shared ancestral reticulation node ancestral to all member co-recombinant isolates. When $|\mathcal{C}| = 1$, it is parameterized as a terminal lineage event.

S3.2. Structural Model Classes

A candidate evolutionary history H is parameterized by an ordered partitioning of the alignment $[0, L)$ into P intervals $\{[s_p, e_p)\}_{p=1}^{P}$ and an associated graph topology:

1. *Clonal Null* (H_0): A single unpartitioned tree across the full locus ($P = 1$), requiring $2N - 3$ branch lengths and 4 substitution parameters ($df = 2N + 1$).
2. *Unpolarized Partitioned Trees* ($H_{\text{partition}}$): Independent unconstrained trees across all P partitions, requiring $df = P(2N - 3) + 4P$ free parameters.
3. *Directed Polarized ARG* (H_{ARG}): A single ancestral backbone tree where reticulations are represented as directed transfer edges that re-attach the recombinant clade inside the transfer interval $[u_1, u_2]$. Each reticulation edge adds only 2 free parameters (source node and target branch attachment) rather than $2N - 3$ unconstrained branch lengths, which yields $df = P_{\text{backbone}}(2N - 3) + 2|\mathcal{E}_{\text{retic}}| + 4$.
4. *Modular Functional Architectures*: Partition models defined by structural protein domains (e.g., Spike NTD, ACE2 receptor-binding domain, and S2 fusion core).

S3.3. Multidimensional Statistical Validation Engine

Competing evolutionary histories are evaluated across four orthogonal statistical criteria:

1. *Phylogenetic Profile Likelihood and Information Criteria*: Across each partition $p \in \{1, \ldots, P\}$, log-likelihood $\ln \mathcal{L}_p = \ln P(\mathbf{Y}_p \mid T_p, \mathbf{Q}, \boldsymbol{\pi})$ is computed via vectorized Felsenstein pruning. Total log-likelihood $\ln \mathcal{L} = \sum_p \ln \mathcal{L}_p$ is penalized under finite-sample corrected AICc and BIC:
$$\text{AICc} = 2\,df - 2\ln \mathcal{L} + \frac{2\,df(df+1)}{L - df - 1}, \qquad \text{BIC} = df \ln L - 2\ln \mathcal{L}. \tag{S11}$$
2. *Residual Non-Tree Manifold Strain*: The expected distance matrix is the length-weighted mixture of local partition patristic distance matrices:
$$\mathbf{D}_{\text{model}} = \sum_{p=1}^{P} \frac{e_p - s_p}{L} \mathbf{D}^{(p)}_{\text{patristic}}. \tag{S12}$$
The residual non-tree distortion matrix $\mathbf{R} = \mathbf{D}_{\text{obs}} - \mathbf{D}_{\text{model}}$ yields the residual strain ratio:
$$\mathcal{S}_{\text{strain}} = \frac{\|\mathbf{R}\|_F}{\|\mathbf{D}_{\text{obs}}\|_F}. \tag{S13}$$
The percentage of non-tree strain explained by history H is $(1 - \mathcal{S}_{\text{strain}}) \times 100\%$.
3. *Spectral SNR over Random Matrix Bulk Edge*: Matrix $\mathbf{R}$ is centered via double-centering $\mathbf{B}_R = -\frac{1}{2}\mathbf{H}(\mathbf{R}^{\odot 2})\mathbf{H}$. The leading eigenvalue $s_1 = \lambda_{\max}(\mathbf{B}_R)$ is compared against the theoretical Marchenko-Pastur bulk noise edge:
$$s_{\text{bulk}} = \sigma^2_{\text{res}} \left(1 + \sqrt{\frac{N}{L}}\right)^2, \tag{S14}$$
where σ_{res} is the standard deviation of off-diagonal elements in $\mathbf{R}$. An empirical SNR $= s_1/s_{\text{bulk}} \leq 1.0$ certifies that residual strain has collapsed into the bulk noise distribution.
4. *Synapomorphic Site Concordance Ratio (SSCR) and Polarity.* At each polymorphic site u where donor D and home H carry distinct nucleotides ($D_u \neq H_u$), recombinant allele R_u is scored as donor-concordant ($R_u = D_u$) or home-concordant ($R_u = H_u$):
$$\text{SSCR}_{\text{in}} = \frac{n_{\text{in}}(R = D)}{n_{\text{in}}(R = D) + n_{\text{in}}(R = H)}, \qquad \text{SSCR}_{\text{flank}} = \frac{n_{\text{flank}}(R = D)}{n_{\text{flank}}(R = D) + n_{\text{flank}}(R = H)}. \tag{S15}$$
Polarization contrast is $\Delta\text{SSCR} = \text{SSCR}_{\text{in}} - \text{SSCR}_{\text{flank}}$. Statistical significance is evaluated via a one-sided Fisher's exact test on $[[n_{\text{in}}(D), n_{\text{in}}(H)], [n_{\text{flank}}(D), n_{\text{flank}}(H)]]$.

S4. Coalescent ARG Simulation Protocol

Genealogies and ancestral recombination graphs (ARGs) were simulated using Hudson's continuous-time coalescent framework with homologous recombination (Hudson, 1983; Posada and Crandall, 2001). For a sample of $N = 10$ contemporary lineages across an alignment of length $L = 1{,}000$ nucleotides, the coalescent process tracks active ancestral segments backward in time from $t = 0$. At any time t, let c denote the number of active ancestral chromosomes, and let g_i denote the genomic span between the leftmost and rightmost ancestral segments carried by lineage i, with total ancestral span $G = \sum_{i=1}^{c} g_i$.

The waiting time Δt to the next evolutionary event is exponentially distributed with total event rate $\lambda_{\text{tot}} = \lambda_{\text{coal}} + \lambda_{\text{rec}}$, where:
$$\lambda_{\text{coal}} = \frac{c(c-1)}{2}, \qquad \lambda_{\text{rec}} = \frac{\rho \cdot G}{2L}. \tag{S16}$$
The population recombination parameter is defined as $\rho = 4N_e r$, where N_e is the effective diploid population size and r is the per-generation recombination probability across the L nucleotide locus. With probability $\lambda_{\text{coal}}/\lambda_{\text{tot}}$, a coalescence event merges two randomly chosen lineages

into an ancestral node. With probability $\lambda_{\text{rec}}/\lambda_{\text{tot}}$, a recombination event selects an eligible lineage with probability proportional to its span g_i and cleaves it at a uniform breakpoint coordinate $b \in [u_{\text{start}}, u_{\text{end}}]$ to partition ancestral material into left and right lineages. Coalescent tracking halts when all L nucleotide sites reach their respective MRCA and yields an ordered sequence of P genealogical partitions, each defined by an interval $[s_p, e_p)$ and a local gene tree T_p.

Nucleotide sequences were evolved along the branch lengths of each local gene tree T_p under the continuous-time Markov HKY85 substitution model (Hasegawa et al., 1985). The model parameterizes unequal base frequencies $\boldsymbol{\pi} = [0.40, 0.20, 0.30, 0.10]$ and transition/transversion bias $\kappa = 2.0$. The infinitesimal generator matrix $\mathbf{Q} = (Q_{ij})$ is defined by:

$$Q_{ij} = \begin{cases} \kappa\mu\pi_j & \text{for transitions } (A \leftrightarrow G, C \leftrightarrow T), \\ \mu\pi_j & \text{for transversions,} \\ -\sum_{k \neq i} Q_{ik} & \text{for } i = j, \end{cases} \tag{S17}$$

normalized such that average substitution rate $-\sum_i \pi_i Q_{ii} = 1.0$. Branch lengths t_b generated in coalescent units were scaled by the population mutation parameter $\theta = 4N_e\mu \in \{10, 50, 100, 200\}$ so that the effective branch length is $b = t_b \cdot \theta/(2L)$. Site-rate heterogeneity was introduced by drawing independent site mutation rate multipliers $r_u \sim \text{Gamma}(\alpha, 1/\alpha)$ across $u \in [0, L)$ such that $\mathbb{E}[r_u] = 1.0$ and $\text{Var}(r_u) = 1/\alpha$ across shape parameters $\alpha \in \{2.0, 0.5, 0.05\}$.

S5. Specialized Epidemiological and Genomic Simulation Suites

Four specialized simulation generators probed operational regimes outside the neutral coalescent (Figure 2):

- *Epidemic Clonal Expansion*: Trees were constructed by generating a bifurcating clonal backbone of 40 background parent taxa across three parental divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, with parental terminal branch drift $\delta = 0.002$). An ancestral recombination event transferred a 1,500 nt cassette (coordinates 1,500–3,000) from Parent 2 to an ancestral recipient in Parent 1 to form recombinant ancestor R_{anc}. From R_{anc}, a burst of $K_{\text{desc}} \in \{2, 5, 8, 10, 15\}$ descendant lineages was generated with short terminal branch divergence ($d_{\text{swarm}} = 0.002$, and 0.005 in the $N = 50$–1,500 large-cohort scaling benchmark) to model transmission swarms in clinical cohorts ($N \in \{42, 45, 48, 50, 55\}$, 20 replicates each, 300 alignments, $L = 5{,}000$ nt, $K_{\text{true}} = 1$).
- *Multi-Way Mosaic Swarms*: Backbones comprised 45 background parent taxa evenly divided among three distinct parental clades (15 taxa each for P1, P2, P3) at $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ across $L = 6{,}000$ nt. Three multi-way topologies were simulated. In the *Dual Cassette* regime, recombinant lineages acquired a 1,200 nt cassette from P2 (coordinates 1,200–2,400) and a separate 1,200 nt cassette from P3 (coordinates 3,600–4,800) embedded within a P1 background ($L = 6{,}000$ nt). In the *Nested Russian Doll* regime, an outer 3,000 nt cassette from P2 (coordinates 1,500–4,500) embedded an inner 1,000 nt cassette from P3 (coordinates 2,500–3,500) to produce a hierarchical $P_1 \to P_2 \to P_3 \to P_2 \to P_1$ mosaic ($L = 6{,}000$ nt). In the *Triple Crossover Cascade* regime, sequential crossovers produced a serial chromosome mosaic composed of P1 (1–2,000 nt), P2 (2,001–4,000 nt), and P3 (4,001–6,000 nt) across $L = 6{,}000$ nt. Descendant swarms spanned $K_{\text{desc}} \in \{1, 5, 10\}$ ($N \in \{46, 50, 55\}$, 20 replicates each, 540 alignments).
- *Ghost Donor Introgression*: Alignments of length $L = 5{,}000$ nt contained 30–40 background parent taxa across three ghost donor regimes ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, $K_{\text{desc}} \in \{2, 5, 10\}$, $N = 32$–50, 20 replicates each, 540 alignments). In the *Surrogate Sister Clade* regime, an ancestral lineage diverged into donor D_{ghost} and surrogate sister P_2' (surrogate divergence $\Delta d_{\text{surr}} = 0.03$, i.e. 0.015 per branch), after which D_{ghost} transferred cassette [1,500, 3,000] to P_1 before omission while sister P_2' remained in the alignment. In the *Pure Ghost Lineage* regime, the donor lineage branched from the tree root and transferred cassette [1,500, 3,000] into P_1 before complete omission so that only unrelated contemporary taxa remained in the sample. In the *Deep Zoonotic Reservoir* regime, the introgressing tract originated from an outgroup lineage with cross-species evolutionary distance $d_{\text{reservoir}} = 0.15$ into P_1.
- *Bacterial Micro-Conversions*: Alignments ($L = 5{,}000$ nt) contained 30 background parent taxa and recombinant swarms ($K_{\text{desc}} \in \{2, 5, 10\}$, $N = 32$–40) across five conversion tract lengths ($L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt, centered at coordinate 2,500) and three divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, 20 replicates each, 900 alignments). The mean number of informative nucleotide substitutions per micro-tract ranged from 0.8–5.2 mutations at 50 nt to 15.1–76.1 mutations at 800 nt.

S6. Grand-Scale Continuous Parameter Sweep (1,000,000 Genomes)

To evaluate algorithmic performance, false-positive rate calibration, and multi-event mosaic resolution across unconstrained continuous evolutionary space rather than isolated grid points, we executed a grand continuous parameter sweep of 1,000,000 independent whole-genome alignments. The sweep comprised 333,334 non-recombinant clonal nulls and 666,666 recombinant chimeras (Supplementary Table S3, Figure 3, and Supplementary Figure S1). Each trial sampled an independent parameter vector uniformly or log-uniformly across evolutionary phase space:

- *Alignment Length (L)*: Sampled log-uniformly from [1,000, 30,000] nt and discretized to multiples of 50 nt ($L \in$ [1,000, 30,000] nt, median 5,450 nt).
- *Sample Taxon Count (N)*: Stratified across three cohort surveillance regimes: 40% small outbreak cohorts ($N \in [4, 30]$ taxa), 40% medium surveillance cohorts ($N \in [31, 100]$ taxa), and 20% large population cohorts ($N \in [101, 250]$ taxa), sampled uniformly within each tier.

- *Parental Divergence (d_{parent})*: Sampled log-uniformly across $d_{\text{parent}} \in [0.005, 0.250]$ substitutions per site.
- *Substitution Dynamics*: Continuous-time Markov substitution under the HKY85 model with transition/transversion ratio $\kappa = 2.5$ and empirical base frequencies $\boldsymbol{\pi} = [0.30, 0.20, 0.20, 0.30]$. Position-specific finite-time transition matrices $\mathbf{P}(t)$ were precomputed analytically per trial for rapid, exact simulation without transcendental overhead.
- *Site Rate Heterogeneity and Biological Heterotachy (α)*:
  - *Strict Molecular Clock* (25% of trials, N = 250,000 overall, 83,311 in null partition): α = 1,000.0 (uniform substitution rates across all sites).
  - *Continuous Gamma Heterotachy* (75% of trials, N = 750,000 overall, 250,023 in null partition): Shape parameter drawn log-uniformly from $\alpha \sim \text{LogUniform}(0.05, 5.00)$, with position-specific relative mutation rates $r_u \sim \text{Gamma}(\alpha, \alpha)$ sampled independently per site and clamped to $[0.01, 50.0]$.
- *Lineage Architecture and Transmission Drift*: The ancestral root sequence evolved into parental nodes P_1, P_2, and optional P_3 (for cohorts with $N \geq 8$ taxa). Contemporary leaves accumulated post-recombination private mutations along terminal branches with transmission drift $d_{\text{drift}} \sim \text{Uniform}(0.0005, 0.0035)$ subs/site.
- *Reticulation Cardinality (K_{true})*:
  - $K_{\text{true}} = 0$ (33.33% of trials, N = 333,334): Strictly non-recombinant clonal nulls to calibrate empirical false-positive rates under severe rate variation.
  - $K_{\text{true}} = 1$ (36.64% of trials, N = 366,396, 55.0% of recombinant partition): Single reticulation event per genome.
  - $K_{\text{true}} = 2$ (16.63% of trials, N = 166,279, 24.9% of recombinants): Dual independent recombination events across segmented partitions.
  - $K_{\text{true}} = 3$ (8.04% of trials, N = 80,391, 12.1% of recombinants): Triple independent recombination events (multi-way mosaicism).
  - $K_{\text{true}} = 4$ (5.36% of trials, N = 53,600, 8.0% of recombinants): Quadruple reticulations that include complex nested architectures.
- *Structural Event Topologies*:
  - $5'$ *Terminal Crossovers* (20% of terminal events): Breakpoint initiates at genome start ($u_1 = 1$) and terminates at u_2 sampled uniformly from 50 nt to the segment bound.
  - $3'$ *Terminal Crossovers* (20% of terminal events): Breakpoint initiates at u_1 sampled uniformly from the segment start to $L-50$ nt and extends to chromosome end ($u_2 = L$).
  - *Intercalary Conversion Cassettes* (60% of events): Internal tracts ($1 < u_1 < u_2 < L$) with tract length L_{tract} sampled log-uniformly from [30 nt, $0.65 \times$ segment length] across all $K_{\text{true}} \in \{1, 2, 3, 4\}$ partitions.
- *Multi-Donor Mosaicism ($P_1 \leftarrow P_2 + P_3$)*: In cohorts with $N \geq 8$ taxa (N = 179,233 trials), recombinant chimeras were generated with multiple distinct donor clades where distinct segments of the same genome were inherited from independent ancestral lineages P_2 and P_3 relative to recipient background P_1.

Simulations were implemented in native compiled Rust (`crates/rhizaeon-cli/src/bin/grand_sweep.rs`) leveraging Rayon data parallelism. The 1,000,000 trials were distributed across cluster infrastructure as a 50-task Slurm array (50 tasks × 20,000 trials, 32 CPU threads and 48 GB RAM per task across 512 distributed CPU cores). The benchmark maintained a sustained throughput of ≈ 380 complete whole-genome simulations and scans per second and completed all 1,000,000 trials in 43.8 minutes. Outputs were aggregated, deduplicated, and verified using `scripts/simulations/merge_sweep_1m.py` to ensure exact, reproducible execution across the entire corpus.

S7. Ground Truth Extraction and Bijective Bipartite Matching

To establish ground truth for reticulation events without relying on heuristic approximations, we computed exact local patristic distance matrices $\mathbf{D}^{(p)} \in \mathbb{R}^{N \times N}$ across every genealogical partition $p \in \{1, \ldots, P\}$ from the branch lengths of tree T_p. For each contemporary taxon i, its local phylogenetic sister at nucleotide $u \in [s_p, e_p)$ is the closest non-identical taxon:

$$\sigma(i, u) = \arg\min_{j \neq i} D_{ij}^{(p)}. \tag{S18}$$

The background Home parent H_i is defined as the majority sister across the full sequence: $H_i = \text{mode}_{u \in [0,L)} \sigma(i, u)$. A true reticulation event occurs when taxon i's sister shifts to a distinct donor lineage $D = \sigma(i, u) \neq H_i$ across a contiguous segment $[u_1, u_2]$. To eliminate trivial non-divergent ancestral coalescences, ground truth events were filtered to require at least one distinguishing mutation between Home and Donor across the tract ($\sum_{u=u_1}^{u_2} \mathbb{I}(mat[H, u] \neq mat[D, u]) \geq 1$).

To score inferred events against ground truth without bias, predicted events $\{\hat{E}_j\}_{j=1}^{\hat{K}}$ were paired to true events $\{E_i\}_{i=1}^{K_{\text{true}}}$ via maximum-weight bijective bipartite matching using the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`). The pairwise affinity matrix $A_{ij} \in [0, 1]$ between true event $i = (C_i, H_i, D_i, u_{1,i}, u_{2,i})$ and inferred event $j = (\hat{C}_j, \hat{H}_j, \hat{D}_j, \hat{u}_{1,j}, \hat{u}_{2,j})$ was formulated as:

$$A_{ij} = S_{\text{triad}}(i, j) \cdot J_{\text{SNP}}(i, j). \tag{S19}$$

Here $S_{\text{triad}} = w(C_i, \hat{C}_j) \cdot \frac{1}{2} \left[w(H_i, \hat{H}_j) + w(D_i, \hat{D}_j) \right]$ weights lineage attribution using tree patristic distances with Gaussian kernel width $\sigma = \bar{D}/2$. Spatial overlap $J_{\text{SNP}} = |S_{\text{true}} \cap S_{\text{inferred}}| / |S_{\text{true}} \cup S_{\text{inferred}}|$ is the Jaccard index evaluated over distinguishing polymorphic sites

between Home and Donor. Unmatched inferred events were penalized as false alarms (reducing precision, PPV $= \hat{K}^{-1} \sum_{\mathcal{M}} A_{ij}$), whereas unmatched true events reduced sensitivity (TPR $= K_{\text{true}}^{-1} \sum_{\mathcal{M}} A_{ij}$).

S8. Empirical Genomic Cohorts and Annotations

Sarbecovirus Whole-Genome Cohort and Coordinate Frame Conversion.

We analyzed two nested whole-genome alignments compiled from GISAID and GenBank. The primary 78-taxon sarbecovirus alignment of Lytras et al. (2022) spans 30,956 bp in multiple sequence alignment (MSA) coordinates with 13,550 total polymorphic columns and 13,160 gap-filtered segregating sites. An expanded 83-taxon alignment (30,965 bp MSA, 13,260 gap-filtered sites) incorporates the five Laotian *Rhinolophus* bat sarbecoviruses (BANAL-20-52, BANAL-20-103, BANAL-20-116, BANAL-20-236, and BANAL-20-247) reported by Temmam et al. (2022). The five Laotian BANAL genomes introduce a 9 nt insertion in the Spike N-terminal domain around 78-MSA position 22,050, which shifts all downstream alignment coordinates by +9 nt relative to the 78-taxon MSA. In their primary study, Lytras et al. (2022) reported open reading frame annotations and the 21 consensus GARD breakpoints (derived from a 22-virus subset) in both ungapped SARS-CoV-2 Wuhan-Hu-1 reference coordinates (NC_045512.2, 29,903 nt, columns `SC2_start`/`SC2_end` in their Supplementary Table S3) and 78-taxon MSA coordinates (columns `al_start`/`al_end`). To prevent cross-frame coordinate discrepancies, Table S2 provides the exact bijective conversion across all three coordinate systems: Wuhan-Hu-1 reference (1–29,903 nt), 78-taxon MSA (1–30,956 nt), and 83-taxon MSA (1–30,965 nt). The table maps all sarbecovirus open reading frames, functional domains, and recombination hotspots and coldspots. It also pairs all 21 published GARD breakpoints with the closest `RhizAeon` continuous changepoints evaluated within each matching frame.

Streptococcus pneumoniae PMEN1 Whole Chromosome.

We analyzed the complete 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 / PMEN1 lineage of *S. pneumoniae* (ATCC 700669, 11 representative global isolates spanning 1984–2008, Croucher et al. 2011). Lossless SNP indexing identified 1,516 segregating polymorphic sites (1,002 parsimony-informative sites) across the 2,221,315 bp chromosome. Annotations follow ATCC 700669 (FM211187.1) across resistance and virulence loci: the *pbp2x* mosaic resistance tract (nt 285,069–305,751, spanning *pbp2x* coding sequence nt 287,466–289,718), capsular polysaccharide locus *cps* (nt 303,559–314,822; extended *pbp2x–cps–pbp1a* recombination hotspot island nt 285,000–352,000), *pbp1a* (nt 323,721–325,108), dihydrofolate reductase *folA* (coding locus nt 1,532,197–1,532,703; trimethoprim resistance recombination island nt 1,620,528–1,649,446), *pbp2b* (nt 1,612,429–1,614,471), and the surface antigen adhesin island *pspC/cbpA* (nt 2,105,000–2,138,000, spanning *pspC* coding locus nt 2,109,335–2,111,878).

Hominoidea MHC Class I Allelic Cohort.

We analyzed $N = 379$ classical MHC Class I alleles (1,128 bp coding alignment, 526 segregating polymorphic sites and 451 parsimony-informative sites across six hominoid species: human, common chimpanzee, bonobo, western gorilla, and Bornean and Sumatran orangutans, Maccari et al. 2017). Alleles were retrieved from the IPD-IMGT/HLA and IPD-MHC databases, comprising 71 human (21 *HLA-A*, 36 *HLA-B*, and 14 *HLA-C*), 162 common chimpanzee (*Patr*), 51 bonobo (*Papa*), 37 western gorilla (*Gogo*), 40 Bornean orangutan (*Popy*), and 18 Sumatran orangutan (*Poab*) alleles. Exon annotations follow canonical classical HLA Class I structure: Exon 1 (Leader peptide, nt 1–72), Exon 2 (α_1 domain, nt 73–342), Exon 3 (α_2 domain, nt 343–618), Exon 4 (α_3 domain, nt 619–894), Exon 5 (Transmembrane anchor, nt 895–1,011), and Exons 6–7 (Cytoplasmic tail, nt 1,012–1,128).

Table S1 Quantitative comparison of `RhizAeon` and 3SEQ across the 3,200-trial Posada and Crandall (2001) benchmark suite. Results are grouped by evolutionary regime: false positive evaluation under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$, $\rho = 0$, 1,200 trials), strict molecular-clock null controls ($\alpha = \infty$, $\rho = 0$, 400 trials), and recombinant coalescent trials ($\rho \in \{1, 4, 16, 64\}$, 1,600 trials). Columns report the population mutation parameter (θ), population recombination parameter (ρ), binary alignment-level power or FPR (%), mean inferred event count ($\hat{K}$), continuous bipartite event precision (PPV), triad lineage attribution accuracy (S_{triad}), and execution runtime per alignment (ms). Whereas alignment-level power scores binary locus detection, PPV weights one-to-one Hungarian-matched events by $S_{\text{triad}} \times J_{\text{SNP}}$ and penalizes ungrounded or over-split calls (Section 3.1).

		`RhizAeon` (Continuous Manifold Tracking)					3SEQ (Discrete Triplet Scanning)				
Evolutionary Regime	Trials	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)
A. Rate Heterogeneity and Clock-Like Null Suites ($\rho = 0$, Empirical False Positive Calibration, Total $N = 1{,}600$)											
$\theta = 10, \alpha \in \{0.05, 0.5, 2.0\}$	300	0.0%	0.00	100.0%	100.0%	22.7	0.0%	0.00	100.0%	100.0%	28.2
$\theta = 50, \alpha \in \{0.05, 0.5, 2.0\}$	300	0.0%	0.00	100.0%	100.0%	16.2	0.0%	0.00	100.0%	100.0%	26.6
$\theta = 100, \alpha \in \{0.05, 0.5, 2.0\}$	300	0.3%	0.00	99.7%	99.7%	15.4	0.3%	0.00	99.7%	99.7%	27.4
$\theta = 200, \alpha \in \{0.05, 0.5, 2.0\}$	300	2.0%	0.02	98.0%	98.0%	15.5	0.3%	0.00	99.7%	99.7%	29.2
All Gamma Rate-Variation Nulls	*1,200*	*0.58%*	*0.01*	*99.4%*	*99.4%*	*17.4*	*0.17%*	*0.00*	*99.8%*	*99.8%*	*27.9*
$\theta = 10, \alpha = \infty$ (Clock Null)	100	0.0%	0.00	100.0%	100.0%	11.9	0.0%	0.00	100.0%	100.0%	59.7
$\theta = 50, \alpha = \infty$ (Clock Null)	100	0.0%	0.00	100.0%	100.0%	13.1	2.0%	0.03	98.0%	98.0%	30.3
$\theta = 100, \alpha = \infty$ (Clock Null)	100	1.0%	0.01	99.0%	99.0%	13.4	0.0%	0.00	100.0%	100.0%	33.4
$\theta = 200, \alpha = \infty$ (Clock Null)	100	2.0%	0.03	98.0%	98.0%	14.4	1.0%	0.06	99.0%	99.0%	36.3
All Clock-Like Nulls ($\alpha = \infty$)	*400*	*0.75%*	*0.01*	*99.2%*	*99.2%*	*13.2*	*0.75%*	*0.02*	*99.2%*	*99.2%*	*39.9*
B. Recombinant Coalescent Suite ($\rho > 0$, Statistical Power and Lineage Attribution, Total $N = 1{,}600$)											
$\theta = 10$ (Low Divergence)	400	8.0%	0.10	20.4%	20.9%	13.2	10.2%	0.26	19.1%	19.7%	31.3
$\rho = 1$	100	3.0%	0.05	54.5%	54.6%	12.1	2.0%	0.04	54.0%	54.2%	41.1
$\rho = 4$	100	9.0%	0.11	22.4%	22.9%	12.0	16.0%	0.40	20.1%	21.2%	37.3
$\rho = 16$	100	18.0%	0.20	3.9%	5.0%	13.8	22.0%	0.58	2.1%	3.2%	23.2
$\rho = 64$	100	2.0%	0.02	1.1%	1.2%	15.0	1.0%	0.03	0.2%	0.3%	23.7
$\theta = 50$ (Moderate Divergence)	400	65.0%	1.27	37.6%	46.5%	18.6	76.8%	6.97	16.6%	29.6%	27.6
$\rho = 1$	100	28.0%	0.36	56.8%	58.5%	14.4	38.0%	2.14	41.0%	48.5%	31.7
$\rho = 4$	100	55.0%	0.79	27.3%	30.7%	16.7	73.0%	5.18	7.0%	17.0%	28.9
$\rho = 16$	100	82.0%	1.63	24.7%	35.8%	20.6	96.0%	9.75	6.3%	19.1%	24.5
$\rho = 64$	100	95.0%	2.29	41.6%	61.3%	22.6	100.0%	10.82	11.9%	33.8%	25.4
$\theta = 100$ (Substantial Divergence)	400	73.5%	1.69	40.3%	49.6%	20.8	84.2%	10.23	14.2%	29.4%	31.2
$\rho = 1$	100	36.0%	0.43	52.4%	54.1%	15.8	52.0%	3.84	34.9%	46.9%	36.7
$\rho = 4$	100	73.0%	1.14	38.2%	44.5%	18.7	85.0%	8.38	6.3%	17.9%	31.8
$\rho = 16$	100	88.0%	1.95	34.6%	45.7%	23.6	100.0%	14.17	5.4%	19.4%	27.8
$\rho = 64$	100	97.0%	3.22	35.8%	54.1%	25.3	100.0%	14.54	10.3%	33.5%	28.4
$\theta = 200$ (High Divergence)	400	74.5%	2.12	43.7%	54.0%	26.2	83.0%	12.12	16.4%	31.4%	36.8
$\rho = 1$	100	32.0%	0.39	63.2%	65.3%	16.7	43.0%	4.05	46.5%	55.1%	39.6
$\rho = 4$	100	72.0%	1.23	36.5%	42.5%	21.5	89.0%	10.92	4.6%	16.7%	37.0
$\rho = 16$	100	97.0%	2.82	37.5%	51.7%	35.8	100.0%	17.25	5.6%	21.2%	37.3
$\rho = 64$	100	97.0%	4.06	37.6%	56.4%	30.7	100.0%	16.26	8.9%	32.6%	33.3

Table S2 Bijective coordinate conversion across SARS-CoV-2 Wuhan-Hu-1 reference coordinates (NC_045512.2, 1–29,903 nt), the 78-taxon multiple sequence alignment of Lytras et al. (2022) (1–30,956 nt), and the expanded 83-taxon multiple sequence alignment incorporating the five Laotian BANAL isolates of Temmam et al. (2022) (1–30,965 nt). Part A lists major open reading frames, Spike functional domains, coldspots, hotspots, and BANAL-247 micro-conversion tracts. Part B lists all 21 consensus GARD breakpoints from Lytras et al. (2022) (Supplementary Table S3) alongside the closest `RhizAeon` changepoints and spatial deviations ($|\Delta|$, nt) evaluated strictly within the matching coordinate frame (83-taxon MSA frame vs. Wuhan-Hu-1 reference frame).

	Genomic Coordinate Frame (nt)			`RhizAeon` Concordance (83-Taxon Cohort)		
Genomic Feature / Breakpoint	Wuhan-Hu-1 Ref (NC_045512.2)	78-Taxon MSA (Lytras et al. 2022)	83-Taxon MSA (+5 Laos BANAL)	`RhizAeon` (83-MSA nt)	`RhizAeon` (Ref nt)	Deviation $\|\Delta\|$ (MSA / Ref nt)
A. Open Reading Frames, Functional Domains, and Recombination Landmarks						
ORF1ab replicase polyprotein	266–21,555	318–21,788	318–21,788	—	—	—
RdRp catalytic core (nsp12 coldspot)	13,442–16,236	13,674–16,468	13,674–16,468	—	—	—
Spike glycoprotein (S)	21,563–25,384	21,796–25,692	21,796–25,701	—	—	—
Spike 5′ NTD hotspot peak	21,566	21,825	21,825	21,825	21,566	—
Spike NTD (codons 13–304)	21,599–22,474	21,858–22,761	21,858–22,770	—	—	—
BANAL-247 NTD micro-conversion 1	21,647–21,749	21,907–22,009	21,909–22,011	21,909–22,011	21,647–21,749	—
BANAL-247 NTD micro-conversion 2	21,791–21,983	22,052–22,243	22,054–22,251	22,054–22,251	21,791–21,983	—
BANAL-247 NTD micro-conversion 3	22,044–22,297	22,310–22,584	22,318–22,592	22,318–22,592	22,044–22,297	—
Spike codon 313 NTD/RBD hinge	22,468	22,755	22,764	22,764	22,468	—
Spike RBD (codons 319–541)	22,517–23,185	22,804–23,481	22,813–23,490	—	—	—
Spike RBD 3′ exit hotspot	23,324	23,620	23,629	23,629	23,324	—
Spike S1/S2 furin cleavage site (codon 685)	23,618	23,923	23,932	—	—	—
Spike S2 fusion core coldspot	24,283–25,283	24,591–25,591	24,600–25,600	—	—	—
ORF3a accessory protein	25,393–26,220	25,701–26,535	25,710–26,544	—	—	—
Envelope (E) / Membrane (M)	26,245–27,191	26,560–27,518	26,569–27,527	—	—	—
ORF6 / ORF7a / ORF7b	27,202–27,887	27,529–28,788	27,538–28,797	—	—	—
ORF8 accessory protein	27,894–28,259	28,795–29,182	28,804–29,191	—	—	—
ORF8 / Nucleocapsid junction hotspot	27,900	28,801	28,810	28,810	27,900	—
Nucleocapsid (N)	28,274–29,533	29,197–30,468	29,206–30,477	—	—	—
B. Published GARD Consensus Breakpoints (Lytras et al. 2022, 22-Virus Subset) vs. `RhizAeon` Changepoints						
GARD BP 1 (nsp2)	1,680	1,732	1,732	1,775	1,723	43 / 43
GARD BP 2 (nsp3 5′)	3,093	3,154	3,154	3,203	3,142	49 / 49
GARD BP 3 (nsp3 macrodomain)	3,649	3,761	3,761	3,807	3,695	46 / 46
GARD BP 4 (nsp3 PLpro)	4,973	5,193	5,193	4,868	4,651	325 / 322
GARD BP 5 (nsp3 3′)	8,208	8,431	8,431	8,483	8,260	52 / 52
GARD BP 6 (nsp6)	11,445	11,671	11,671	11,030	10,804	641 / 641
GARD BP 7 (RdRp 5′ entry)	12,622	12,854	12,854	13,001	12,769	147 / 147
GARD BP 8 (RdRp internal 1)	14,401	14,633	14,633	14,807	14,575	174 / 174
GARD BP 9 (RdRp internal 2)	15,954	16,186	16,186	14,807	14,575	1,379 / 1,379
GARD BP 10 (nsp13 helicase)	16,923	17,155	17,155	18,299	18,067	1,144 / 1,144
GARD BP 11 (nsp15 endoribonuclease)	19,965	20,197	20,197	19,850	19,618	347 / 347
GARD BP 12 (nsp15/nsp16)	20,518	20,750	20,750	20,651	20,419	99 / 99
GARD BP 13 (nsp16 3′)	21,198	21,430	21,430	21,539	21,307	109 / 109
GARD BP 14 (Spike 5′ upstream)	21,411	21,643	21,643	21,621	21,389	22 / 22
GARD BP 15 (Spike NTD/RBD hinge)	22,460	22,747	22,756	22,649	22,353	107 / 107
GARD BP 16 (Spike RBD 3′ exit)	23,396	23,692	23,701	23,629	23,324	72 / 72
GARD BP 17 (Spike S2 upstream)	24,144	24,452	24,461	24,484	24,167	23 / 23
GARD BP 18 (Spike S2 fusion exit)	24,843	25,151	25,160	25,183	24,866	23 / 23
GARD BP 19 (Envelope / Membrane)	26,323	26,638	26,647	26,953	26,604	306 / 281
GARD BP 20 (ORF7a / ORF7b)[a]	27,388	28,165	28,174	27,718	27,369	456 / 19
GARD BP 21 (ORF7b / ORF8)	27,685	28,571	28,580	28,670	27,775	90 / 90
Summary Concordance ($\leq$ 150 nt)	—	—	—	*13 / 21 (61.9%)*	*14 / 21 (66.7%)*	*Median: 107.0 / 99.0*

[a] Between 83-MSA positions 27,718 (Ref 27,369) and 28,174 (Ref 27,388), SARS-CoV-2 Wuhan-Hu-1 contains accessory ORF7a/ORF7b deletions relative to bat sarbecovirus sequences, so the two breakpoints are separated by 19 nt in ungapped Wuhan-Hu-1 reference coordinates versus 456 nt in gapped MSA coordinates.

Table S3 Empirical operational phase envelope of `RhizAeon` across 1,000,000 independent continuous simulation trials. Simulations span $L \in [1{,}000, 30{,}000]$ nt, $N \in [4, 250]$ taxa, $d_{\text{parent}} \in [0.005, 0.25]$ subs/site, and continuous Gamma rate variation $\alpha \in [0.05, 5.0]$ alongside strict molecular clock controls ($\alpha \geq 50$). Multi-event topologies include true nulls ($K_{\text{true}} = 0$), $5'/3'$ terminal crossovers, intercalary conversion cassettes ($K_{\text{true}} = 1$), and multi-donor / multi-event mosaicism ($K_{\text{true}} \in \{2, 3, 4\}$). Breakpoint errors report median absolute error (nt) among detected recombinant tracts.

Evolutionary Regime	Parameter Space	Total Trials	Detected / Rec	Power / FPR	Mean Jaccard	Median BP Error
Strict Molecular Clock Null	$d_{\text{parent}} \in [0.005, 0.25], \alpha \geq 50$	83,311	761	0.913% FPR	—	—
Gamma Heterotachy Null	$\alpha \in [0.05, 5.0], d_{\text{parent}} \in [0.005, 0.25]$	250,023	2,593	1.037% FPR	—	—
Taxon-Level False Alarm	$N_{\text{taxa}} = 22{,}744{,}905$ total leaves	333,334	5,884 taxa	0.026% FPR	—	—
Recombinant Partition Stratified by Informative Tract Mutations (M_{tract})						
Tract SNPs $M_{\text{tract}} \in [0\text{–}2]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	85,564	484	0.6%	0.000	42.0 nt
Tract SNPs $M_{\text{tract}} \in [3\text{–}5]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	68,609	4,801	7.0%	0.043	16.0 nt
Tract SNPs $M_{\text{tract}} \in [6\text{–}10]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	77,568	25,024	32.3%	0.238	19.0 nt
Tract SNPs $M_{\text{tract}} \in [11\text{–}15]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	52,607	29,679	56.4%	0.443	19.0 nt
Tract SNPs $M_{\text{tract}} \in [16\text{–}25]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	71,427	52,042	72.9%	0.592	16.0 nt
Tract SNPs $M_{\text{tract}} \in [26\text{–}50]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	96,694	80,562	83.3%	0.703	12.0 nt
Tract SNPs $M_{\text{tract}} \in [51\text{–}100]$	$L_{\text{tract}} \in [30, 0.65L]$ nt	84,676	73,443	86.7%	0.742	10.0 nt
Tract SNPs $M_{\text{tract}} > 100$	$L_{\text{tract}} \in [30, 0.65L]$ nt	129,521	114,518	88.4%	0.728	7.0 nt
Multi-Event and Structural Topology Decomposition ($K_{\text{true}} \geq 1$, Total $N = 666{,}666$)						
Single Event ($K_{\text{true}} = 1$)	Crossover / Cassette	366,396	197,742	54.0%	0.409	14.0 nt
Dual Events ($K_{\text{true}} = 2$)	Two Reticulations	166,279	101,156	60.8%	0.512	10.0 nt
Triple Events ($K_{\text{true}} = 3$)	Three Reticulations	80,391	49,286	61.3%	0.510	8.0 nt
Quad Events ($K_{\text{true}} = 4$)	Complex Mosaicism	53,600	32,369	60.4%	0.492	8.0 nt
Terminal Crossovers	$5'$ or $3'$ Exchange	239,767	163,424	68.2%	0.514	10.0 nt
Internal Cassettes Only	Intercalary Swap	426,899	217,129	50.9%	0.420	12.0 nt

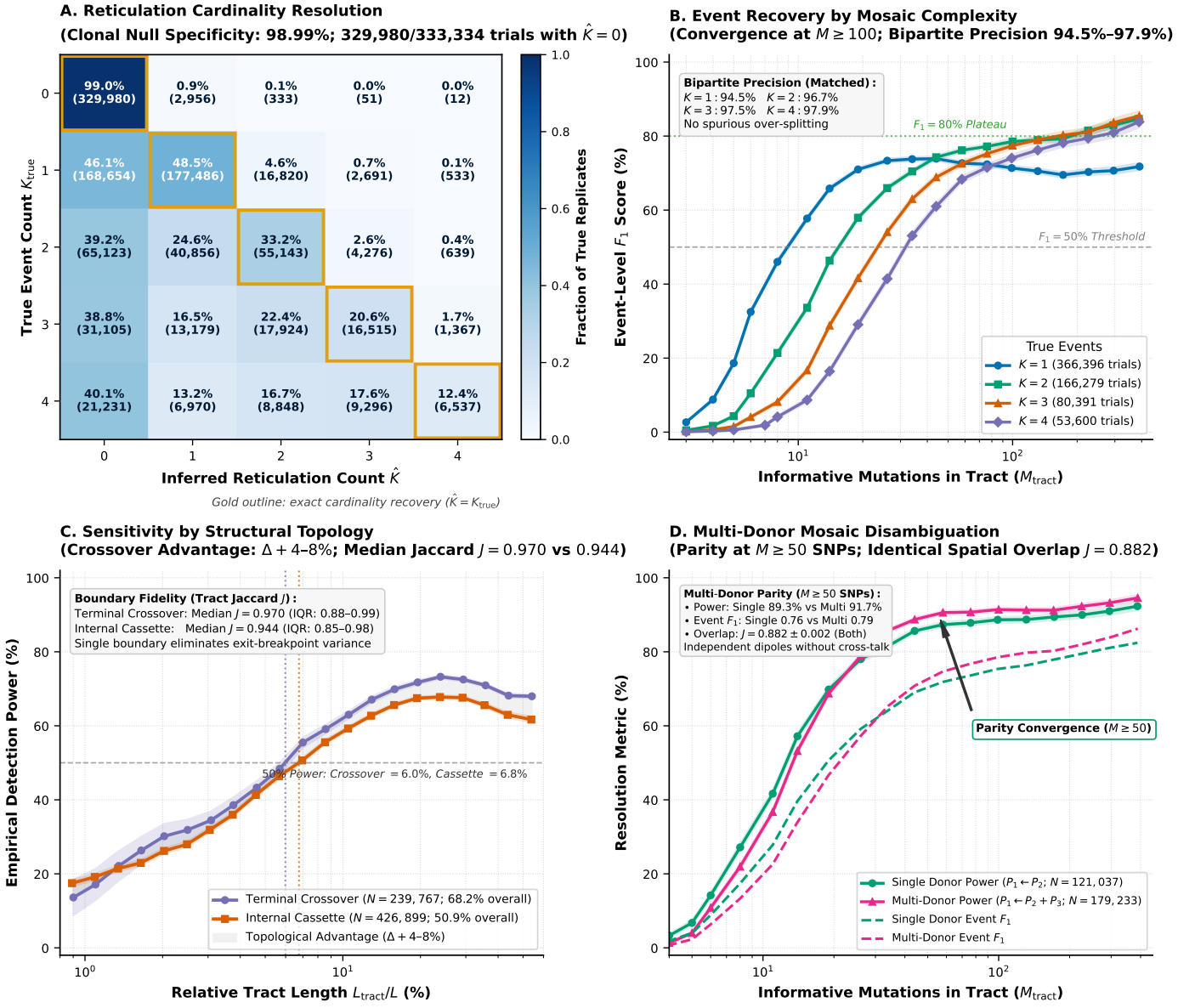


Figure S1 Multi-event reticulation resolution, structural topology sensitivity, and multi-donor disambiguation across the 1,000,000-trial simulation sweep. (A) Cardinality confusion matrix comparing true (K_{true}) versus inferred ($\hat{K}$) event counts across 333,334 clonal nulls ($K_{\text{true}} = 0$, 98.99% specificity, 1.01% alignment-level FPR) and 666,666 recombinant cohorts ($K_{\text{true}} \in \{1, 2, 3, 4\}$). (B) Event-level F_1 score across informative tract mutations (M_{tract}) stratified by $K_{\text{true}} \in \{1, 2, 3, 4\}$ with 95% bootstrap confidence ribbons. (C) Detection power across relative tract length (L_{tract}/L) comparing terminal crossovers ($N = 239{,}767$) against intercalary cassettes ($N = 426{,}899$). (D) Sensitivity and event F_1 score across M_{tract} comparing single-donor ($P_1 \leftarrow P_2$) versus multi-donor ($P_1 \leftarrow P_2 + P_3$) mosaics.